

Total design load W_c in kilograms per metre of pipe length for cover depth H in metres

$H=$ 3.7	4.0	4.3	4.6	4.9	5.2	5.5	5.8	6.1	6.4	6.7	7.0	7.3	7.6
1448	1566	1683	1801	1918	2036	2154	2271	2389	2506	2624	2742	2859	2977
278	250	226	205	186	170	156	143	132	122	113	105	98	92
97	83	72	63	56	49	44	39	35	32	29	27	24	22
54	47	41	36	32	28	25	23	21	19	17	16	14	13
2260	2340	2390	2450	2500	2540	2580	2620	2640	2660	2690	2710	2730	2750
2107	2278	2449	2620	2791	2963	3134	3305	3476	3648	3819	3990	4161	4333
399	359	324	293	267	244	224	206	189	175	162	151	140	131
137	118	102	90	79	70	62	56	50	46	41	38	35	32
78	67	59	52	46	41	36	33	30	27	24	22	20	19
2260	2340	2390	2450	2500	2540	2580	2620	2640	2660	2690	2710	2730	2750
2407	2602	2798	2994	3189	3385	3581	3776	3972	4168	4364	4559	4755	4951
454	408	369	334	304	278	254	234	216	199	185	171	160	149
156	134	116	102	90	80	71	64	57	52	47	43	39	36
89	77	67	59	52	46	41	37	34	31	28	25	23	21
2710	2810	2900	2970	3040	3100	3160	3210	3260	3300	3330	3360	3390	3400
2683	2902	3120	3338	3556	3775	3993	4211	4430	4648	4866	5084	5303	5521
505	454	410	372	338	309	283	260	240	221	205	191	177	165
173	149	129	113	100	88	79	71	64	58	52	48	44	40
99	86	75	65	58	52	46	41	37	34	31	28	26	24
2710	2810	2900	2970	3040	3100	3160	3210	3260	3300	3330	3360	3390	3400
3006	3251	3495	3740	3985	4229	4474	4718	4963	5208	5452	5697	5942	6186
565	508	458	415	378	345	316	290	268	247	229	213	196	185
193	166	144	126	111	99	88	79	71	64	58	53	49	45
111	96	83	73	65	58	52	46	42	38	35	32	29	27
2710	2810	2900	2970	3040	3100	3160	3210	3260	3300	3330	3360	3390	3400
3328	3599	3879	4141	4412	4583	4954	5225	5496	5767	6038	6309	6580	6851
624	561	506	459	418	381	349	321	296	273	253	235	219	204
213	183	159	139	123	109	97	87	78	71	64	59	54	49
123	106	92	81	72	64	57	51	46	42	38	35	32	29
3180	3300	3420	3520	3610	3700	3780	3830	3910	3950	4010	4050	4090	4130
3915	4234	4553	4872	5191	5510	5829	6148	6467	6786	7105	7424	7743	8061
733	658	594	538	490	447	410	376	347	321	297	276	257	239
249	214	186	163	144	127	114	102	92	83	75	69	63	58
144	124	108	95	84	75	67	60	54	49	45	41	38	34
4650	4870	5080	5260	5440	5600	5750	5900	6020	6140	6250	6340	6440	6500
4641	5019	5397	5775	6153	6532	6910	7288	7666	8044	8422	8801	9179	9575
912	819	739	670	609	556	509	468	431	398	369	343	319	297
308	266	231	202	178	158	141	126	114	103	94	85	78	72
179	155	135	118	105	93	83	75	67	61	56	51	47	43
5160	5410	5650	5870	6080	6270	6440	6610	6760	6900	7040	7150	7260	7380
5524	5975	6425	6876	7326	7777	8228	8678	9129	9579	10,030	10,480	10,931	11,382
1084	974	878	796	724	661	605	556	512	473	438	407	379	353
366	315	274	240	211	188	167	150	135	122	111	101	93	85
213	184	160	141	124	111	99	89	80	73	66	60	55	51
5660	5950	6230	6490	6720	6950	7150	7340	7530	7700	7850	8000	8140	8270
6427	6952	7477	8002	8526	9051	9576	10,101	10,626	11,150	11,675	12,200	12,725	13,250
1260	1132	1021	926	842	768	704	647	596	550	510	473	440	410
425	366	318	278	246	218	194	174	157	142	129	118	108	99
248	214	187	164	145	129	115	103	93	85	77	70	64	59
6190	6500	6820	7100	7390	7630	7860	8090	8300	8500	8680	8870	9020	9170
7340	7940	8539	9139	9739	10,339	10,939	11,539	12,139	12,739	13,339	13,938	14,538	15,138
1439	1292	1166	1057	961	877	803	738	680	628	582	540	502	468
485	417	363	318	280	248	222	199	179	162	147	134	123	113
284	245	213	187	165	147	131	118	106	96	88	80	73	68
7230	7630	8010	8360	8720	9040	9340	9640	9900	10,200	10,400	10,600	10,800	11,000
8261	8937	9613	10,289	10,965	11,540	12,316	12,992	13,668	14,344	15,020	15,696	16,372	17,048
1620	1455	1313	1189	1082	987	904	831	765	707	654	607	565	527
545	469	408	357	315	279	249	224	202	182	166	151	138	127
319	276	240	211	186	165	148	133	120	109	99	90	83	76
7750	8190	8620	9030	9400	9760	10,100	10,400	10,700	11,000	11,300	11,500	11,800	12,000
9137	9885	10,634	11,382	12,130	12,879	13,627	14,375	15,124	15,872	16,620	17,369	18,117	18,865
1792	1609	1452	1316	1197	1092	1000	919	846	782	724	672	625	583
603	519	415	395	348	309	276	247	223	202	183	167	153	141
353	305	266	233	206	183	163	147	133	120	109	100	91	84

Table 5.*

Nominal diameter (mm)	Equivalent load (kgf/m)
600	210
675	270
750	330
825	400
900	490
975	560
1050	650

*When a pipe is running full, its contents are equivalent to an external load of 75% of the weight of water in the pipe.

Buried flexible pipes

The following are guidelines for calculations involving buried flexible pipes.

Notation

Symbol	Definition	Unit
D_1	Deflection lag factor	—
d	Mean diameter of pipe	mm
δ	Reduction in vertical diameter	mm
E_s	Elastic modulus for soil as determined at overburden pressure in tri-axial test	N/mm ²
El/d^3	Stiffness factor for pipe	N/m ²
e	Subgrade modulus	N/mm ² /mm
E_b	Bending strain	—
E_t	Tensile strain	—
H	Height of cover above pipe spring line	m
k	Deflection coefficient dependent upon bedding angle	—
k_s	Meyerhof's constant for granular materials (taken as 1.63 N/mm ² /m depth)	—
P_o	Total external pressure on pipe	N/mm ²
P_{ob}	External pressure on pipe used in buckling strength calculations	N/mm ²
P_{od}	External pressure on pipe used in deflection calculations	N/mm ²
P_c	Critical buckling pressure	N/mm ²
P_e	External pressure on pipe due to backfill	N/mm ²
P_i	Internal pressure	N/mm ²
P_s	External pressure on pipe due to surcharge loading	N/mm ²

P_t	External pressure on pipe due to traffic loading	N/mm^2
P_v	Internal vacuum pressure	N/mm^2
t	Pipe-wall thickness	mm

Loads

Traffic loads (P_t)

Traffic loading may be taken from the appropriate charts, e.g. as given in NBS Special Report No. 37.

Table 6. Turnall pipes: size range and classification

Nominal diameter	Class L		Class M		Class H	
	Minimum ultimate crushing load (W_T)		Minimum ultimate crushing load (W_T)		Minimum ultimate crushing load (W_T)	
mm	kN/m	kfg/m	kN/m	kfg/m	kN/m	kfg/m
100	—	—	—	—	37.95	3870
150	—	—	—	—	37.95	3870
175	—	—	—	—	37.95	3870
200	—	—	—	—	37.95	3870
225	—	—	—	—	37.95	3870
250	—	—	—	—	37.95	3870
300	—	—	35.00	3570	46.68	4760
375	—	—	38.63	3940	52.56	5360
450	—	—	43.73	4460	58.35	5950
525	37.95	3870	47.46	4840	65.70	6700
600	42.16	4300	56.87	5800	77.47	7900
675	48.05	4900	64.23	6550	86.10	8780
750	51.00	5200	68.64	7000	91.88	9370
825	55.40	5650	74.53	7600	100.71	10,270
900	58.35	5950	85.57	8930	107.87	11,000
975	62.76	6400	94.82	9670	116.70	11,900
1050	65.70	6700	99.24	10,120	124.05	12,650

Table 7. Depth h_1 (mm) corresponding to bedding angle α

Nominal diameter (mm)	200	225	250	300	350	400	450	500	600	700	750
Bedding angle α	h_1 (mm)										
60°	←————→	25	←————→	←————→	50	←————→	←————→	75	←————→		
90°	←————→	50	←————→	←————→	100	←————→	←————→	150	←————→		
120°	←————→	100	←————→	←————→	200	←————→	←————→	250	←————→		

For main road traffic loading, normally use either HA or 45 units of HB loading in accordance with BS 153: Part 3A, depending upon the quantity and type of vehicles expected.

Surcharge loads (P_s)

Include here long-term loads.

Backfill (P_e)

Normally use the full weight of the soil above the crown of the pipe. For buckling resistance, where ground-water level is above the pipe, make allowance for buoyancy and add water pressure at crown (at invert if pipe can be empty).

Vacuum (P_v)

Include this if there is a possibility of full or partial vacuum inside the pipe.

Internal pressure (p_i)

This is used in strength calculations to assess the quantity of glass reinforcement necessary to resist bursting. Normally use maximum working pressure including an allowance for surges, and check using test pressure with a reduced factor of safety.

Total external pressure on pipe (P_o)

For deflection calculations $P_{od} = 2P_t + P_s + P_e$.

For strength calculations $P_{ob} = P_t + P_s + P_e + P_v$.

Note: Transient loads, e.g. traffic loads, have less effect in deflecting a flexible pipe than do permanent soil loads and surcharges. Therefore, when considering deflection only, a reduction factor of 0.5 may be applied to P_t .

Deflection

Spangler's formula for the deflection of flexible pipes can be written thus:

$$\frac{\delta}{d} = D_1 \frac{K \times P_{od}}{8EI} \frac{1}{10^6 \times d^3} + 90.031 \text{ e.d.}$$

where δ = reduction in vertical diameter

d = mean diameter of pipe

e = subgrade modulus

$$\frac{EI}{d^3} = \text{pipe stiffness factor}$$

D_1 = deflection lag factor

This is introduced to make allowances for the slow increase in deformation of some soils under sustained lateral pressure. Typical values are in the range 1.0–1.5 for non-pressure pipes. For pressure pipes where internal pressure just balances external load, use $D_1 = 1$. For pipes where high internal pressure is likely to cause re-rounding use values between 0.25 and 1, typically 0.5.

$k = 0.100$ for 60 bedding angle

Meyerhof and Fisher, based on Terzaghi, have shown that:

$$e = \frac{E_s}{0.75 \times d}$$

where E_s = modulus of elasticity of the soil at overburden pressure, as determined in a tri-axial test. If no test results are available, use 4000–10,000 kN/mm² in good ground above water table, according to compaction; 2000 to 5000 kN/m² in poor ground, or in good ground below water table.

Alternatively, in good ground E_s may be assessed by the following relationships:

$E_s = k_s H$ × if backfill is dry granular material

$E_s = k_s H \times \frac{1.7}{2.7}$ if backfill is saturated granular material

where k_s (Meyerhof) is taken in the range 1.09–3.25 N/mm²/m depth, a value of 1.63 N/mm²/m depth being commonly used.

H = height of cover above pipe spring line

With flexible pipes, the effect of the pipe stiffness on the deformation of the composite pipe/soil system is small and can usually be neglected. The Spangler formula then reduces to:

$$\frac{\delta}{d} = D_1 \frac{2.4P_{od}}{E_s}$$

The long-term deflection calculated as previously should not exceed 5% of the diameter. If necessary, better quality surround material, and/or higher compaction should be specified. Allowable initial deflection will depend upon the engineer's assessment of D_1 but, as a guide, this should be limited to a maximum of 3%.

Buckling strength

Based on Meyerhof and Baikie, the critical buckling pressure on a buried pipe (P_c) is given by the following equation:

$$P_c = 4.6 \sqrt{E_s \times \frac{EI}{d^3} \times \frac{1}{10^6}}$$

Therefore, permissible buckling pressure (P_{ob}) is as follows:

$$P_{ob} = \sqrt{\frac{1}{FS} \times 4.6 E_s \times \frac{EI}{d^3} \times \frac{1}{Cf \times 10^6}}$$

where FS = factor of safety against buckling for which a value of 2 is used

Cf = creep factor typically in the range 2–3 depending upon the long-term elastic properties of the pipe-wall material

$\frac{EI}{d^3}$ = pipe stiffness factor (initial)

Bursting strength

Bursting is resisted by glass-reinforced layers acting in hoop tension. The thickness of glass-reinforced layers provided is such that at working pressure there is an initial factor of safety against bursting of between 6 and 7. At test pressure, a factor of safety of at least 4 is normally available.

Crushing strength

Certain pipes have a crushing strength in excess of 80 MN/m². Crushing stress is not normally critical, though.

Longitudinal strength

Pipes should have longitudinal strength in excess of the requirements of BS 5480 Part 1: 1977.

Strain

Bending strain E_b

If the pipe deformed as an eclipse, the strain in the pipe wall due to bending would be:

$$E_b = \frac{3\delta}{d} \times \frac{t}{d}$$

where t = thickness of pipe wall.

Tests by Molin showed that bedding irregularities caused deviations from this theoretical value which were dependent upon pipe stiffness, i.e.:

$$E_b = d \times \frac{\delta}{d} \times \frac{t}{d}$$

where $3 \leq d \leq 6$: d tends towards 3 for stiffer pipes which are less affected by bedding irregularities and hence, for a particular deflection, are subject to correspondingly lower levels of bending strain.

In non-pressure pipes, bending strain should be limited to 0.35%. For pressure pipes, bending strain is generally limited to 0.2%.

Circumferential tensile strain E_t

Circumferential tensile strain in pressure pipes should normally be limited to an initial maximum value of 0.2% at working pressure.

Note: In pressure pipe design, the effects of bending and internal pressure are not additive and may, therefore, be considered separately.

Summary of design criteria

Long-term deflection shall be limited by the allowable bending strains listed or to 5%, whichever is the lesser.

Bending strain, 0.35% (non-pressure); 0.20% (pressure).

Circumferential tensile strain, 0.2%.

Trenching

The preparation of the trench bottom to give an even bedding for the barrel of the pipe, and proper alignment of pipes, is of primary importance. In rocky ground the trench should be extended to at least 100 mm deeper than required and then made up to the required level by introduction of well rammed compactible material of a type which will not be washed away; alternatively, the pipe may be embedded in a layer of freshly mixed concrete. The trench should not normally be opened up more than a few pipe lengths ahead of the point where laying is taking place.

Width of trench

The trench width will depend to some extent on the ground conditions and depth of laying but should be kept to the practical minimum. Minimum widths for normal conditions, as used in the preparation of external loading charts, are based on standard bucket sizes wherever possible and give at least 150 and

200 mm clearances on each side of the pipe for diameters in the ranges 200–350 and 400–750 mm, respectively.

In narrow-trench conditions, the backfill load is a function of the depth of cover and width of trench (B) as measured at the level of the crown of the pipe. Thus the backfill loads will be the same for any of the trench sections shown in Figure 2.

Where a particular width has been specified, it should not be significantly exceeded without reference to the pipeline designer.

Depth of trench

The depths at which a pipeline is to be laid will normally have been determined by the pipeline designer to whom reference should be made if, for any reason, the depths specified for a particular application require to be exceeded by a significant amount. The minimum depth of cover will depend on considerations such as frost, traffic loadings, size and class of pipeline and, as regards adequacy of anchorage, the type and compatibility of the fill material.

Preparation of trench bedding

Class C bedding

In Class C bedding where the pipe barrel is to be in more or less continuous contact with the foundation soil, the initial excavation should be marginally less than the required depth in order that final levelling and preparation of the bedding may be carried out manually. Any high spots should be removed. If overdigging accidentally occurs, the correct level must be regained by introduction of selected material, well-compacted. This preparatory work can be aided by use of a wooden straight-edge, of length not less than that of the

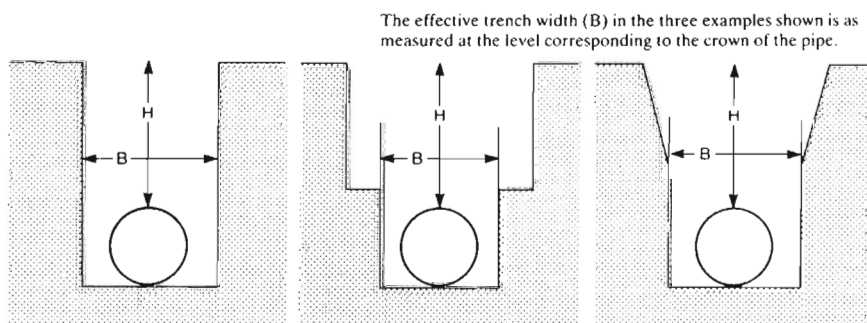


Figure 2.

pipes. One end of the straight edge should be notched. Allowance can be made for slight initial settlement due to the weight of the pipe.

Class B bedding

In Class B bedding it is necessary to overdig the trench by an amount equivalent to the required depth of granular overlay between the pipe and the foundation ground. The straight-edge technique can again be used to obtain the gradient and approximate level of the granular bedding.

For both Class B and Class C bedding it is necessary to excavate a joint hole of sufficient length and depth to give clearance for the coupling to ensure that the pipes rest on the trench bed and not on the coupling.

Backfilling

Backfilling should be carried out in accordance with the requirements specified for the particular pipeline and can take place as soon as the joints have been made. If it is desired to leave joints uncovered until completion of pressure testing, the trench should be backfilled only over the barrel portion of each pipe. In such cases it is necessary to ensure that an adequate depth of compacted backfill is applied over the crown of the pipe to prevent the pipeline lifting when the test pressure is applied.

For Class C bedding selected backfill is compacted to a suitable depth h_1 , which corresponds to the bedding angles as selected for the particular nominal diameter of pipe.

The backfill between levels is ordinary soil, free from lumps and large stones. Compaction will normally be required and should be carried out in stages with layers of 150–300 mm in thickness. The remainder of the excavated material can be used for the top level of backfill, the extent of compaction required depending on local factors.

For Class B bedding the pipe will have been laid on a prepared granular bed, which should not be less than 100 mm for all sizes of pipe where laid in uniform soils. Where laid in rock or mixed soils containing rock bands, boulders, large stones or other irregular hard spots, dimension h should not be less than 200 mm.

To complete the granular bedding, a further layer of suitable depth (according to pipe diameter and bedding angle (α) as selected) should be added and compacted.

Selected backfill, free from lumps or large stones, is then compacted in layers of 150–300 mm to a height of 300 mm above the crown of the pipe. Ordinary backfill material can be used for the remainder, the extent of compaction depending on local requirements.

Joint holes should be backfilled with selected material and properly compacted.

Large stones should be kept well away from the sides of the trench to avoid the possibility of their being accidentally dropped on the pipes which have been laid.

Thermoplastic pipes

CPVC piping is usually selected for its higher temperature characteristics, i.e. 48–94°C (120–200°F). Expansion and contraction could become excessive at the higher ranges with intermittent flow in buried lines.

Expansion joints are recommended for use in suitable pits for the upper temperature limits. In this case, the line should not be 'snaked'. Snaking the pipe with as many loops as possible per 30 m (100 ft) should prove satisfactory for normal conditions for PVC and for temperature ranges up to 60°C (140°F) for CPVC. Sun heat or hot water flow to bring the pipe to 38–44°C (100–110°F) surface temperature is recommended prior to preliminary backfill of the snaked line.

Care should be used to make the best possible solvent-cement joint. Threading should be avoided.

Where the thermoplastic pipe is joined to metal, use of a metal flange with a flexible gasket is suggested.

Cement cure times in excess of those normally recommended are suggested prior to running at the elevated temperature. When thermoplastic pipe is installed underground in trenches, the trench bottom should be smooth and free of rocks and debris.

Trenches should never be used as repositories for rubbish. If the trench is dug in ledge rock, hard pan or where boulders and rocks are not removed, the trench bottom must be padded out with sand or compacted with fine-grain soils.

The trench should be wide enough to provide adequate room for: (1) joining the pipe in the trench, (2) snaking the pipe from side to side in the trench to provide slack for future contraction or expansion, and (3) placing and compacting side fills.

Trench width may be minimised by joining the pipe outside of the trench and lowering it into the trench with levelling supports.

Trench depth is determined by intended service, national standards and recommendations as well as local conditions.

Thermoplastic pipe should be installed at least below frost level. Pipe for conveying liquids susceptible to freezing should be buried no less than 300 mm (12 in) below the maximum frost level. Permanent lines subjected to heavy traffic should have a minimum cover of 600 mm (24 in). For light traffic 300–450 mm (12–18 in) is normally sufficient for small diameter and small diameter-to-thickness ratio pipe. With larger sizes or larger diameter-to-thickness ratio pipe, bearing stresses should be calculated to determine cover required.

As well as local and national codes, reliability and safety should always be paramount.

Bedding and backfilling thermoplastic pipe

With a smooth uniform trench bottom, the pipe will be supported over its entire length on firm, stable material. Blocking should never be used to change pipe grade or to provide intermittent support over low sections in the trench. Because subsoil conditions vary greatly throughout the world, different pipe-bedding problems will be encountered in various localities. In general, subsoil should be stable and should provide physical protection for the pipe.

The pipe should be surrounded with backfill materials having a particle size of 13 mm ($1/2$ in) or less. Backfilling should be carried out in layers with each layer compacted sufficiently so that lateral pressure soil forces are developed uniformly. Under certain conditions, it may be advisable to have the pipe under pressure during the backfilling operation.

When compacting sand or gravels, vibratory methods are recommended. If water flooding is used, the initial backfill should be sufficient to ensure complete coverage of the pipe. Additional backfill should not be added until the water-flooded backfill is firm enough to walk on. Precautions should be taken to stop the pipe floating. In all instances, the trench should be filled completely, and rolling equipment or heavy tampers should be used only to consolidate the final backfill.

With regard to thermoplastic pipe, reference should also be made to ASTM D2774 'Underground Installation of Thermoplastic Pressure Piping' and ASTM D2321 'Underground Installation of Flexible Thermoplastic Sewer Pipe'. See also the Chapter on Thermoplastic Pipe.

Collapsing Pressure for Pipes and Tubes

In the general case of pipes where the length is eight or more times the diameter, the uniformly applied pressure to produce collapse is given by:

$$P = \frac{2E}{1 - \sigma^2} \left(\frac{t}{D} \right)^3$$

where P = external pressure

E = modulus of elasticity of pipe material

σ = Poisson's ratio for pipe material

t = pipe-wall thickness

D = outside diameter

The upper limit for the collapsing pressure is given when the compressive stress produced is equal to the compressive yield strength of the pipe material.

Buried pipes

Buried pipes are subject to both internal and external loading, the general theory stating that the magnitude of internal pressure which a rigid pipe can withstand varies inversely with the magnitude of simultaneously applied external loading. The net effect on the combined load-bearing strength of the pipe can be determined mathematically from the Schlick formula, as follows:

$$P_1 = P_2 \left[1 - \left(\frac{F_1}{W_T} \right)^2 \right]$$

which may also be written as:

$$F_1 = W_T \sqrt{1 - \frac{P_1}{P_2}}$$

- where P_1 = internal hydrostatic pressure (kN/m^2) which will fracture the pipe when acting in combination with some external load F , applied in two-edge bearing
- P_2 = internal hydrostatic pressure (kN/m^2) which will burst the pipe in the absence of any external load
- F_1 = external load (kN/m) applied in two-edge bearing which will fracture the pipe when combined with some internal pressure P_1
- W_T = two-edge bearing load (kN/m) which will crush the pipe in the absence of any internal pressure.

Figure 1 shows a 'combined loading' curve for a pipe which would burst at some internal pressure, P_2 , or crush under some external load, W_T , if either were acting alone. If, however, some lesser internal pressure, P_1 , is acting on the pipe in combination with an external loading, F_1 , the magnitude of F_1 at which fracture will occur can be determined by means of ordinates drawn to intersect at a point X on the curve.

Combined loading

The potential working envelope of a buried pipe is represented by a curve describing maximum internal pressure combined with a curve describing maximum external load (crushing pressure). Thus, because of the interdependence of the two parameters, boundaries are curves rather than straight lines and take the form shown in Figure 2.

Practical curves of this form, known as combined loading charts, can be devised for each particular size and class of pipe and incorporate suitable safety factors against bursting or crushing. Such a chart will indicate maximum working limits, e.g. for any given pressure limit a corresponding safe working crushing strength, or safe working pressure limit.

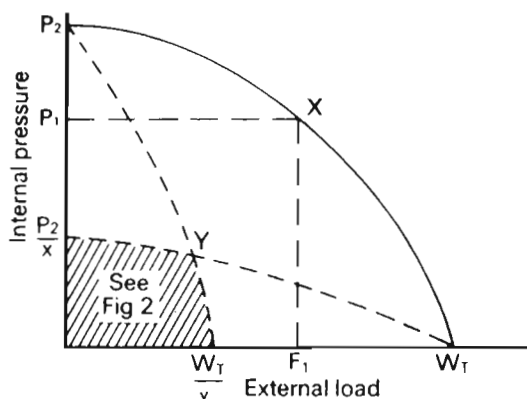


Figure 1.

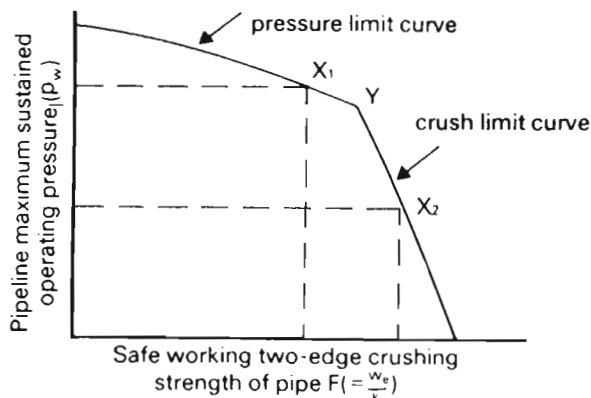


Figure 2.

Basically, such loading charts are modified Schlick curves, taking into account suitable safety factors, e.g.:

$$p_w = \frac{P_2}{X} \left[1 - \left(\frac{F_1}{W_T} \right)^2 \right]$$

$$F = \frac{W_T}{Y} \sqrt{1 - \frac{P_1}{P_2}}$$

- where P_w = maximum sustained operating (or static) pressure (kN/m^2)
 X = factor of safety against bursting when an internal pressure, p_w , is applied together with an external load (see Table 1)
 F_1 = external load (kN/m) applied in two-edge bearing which will fracture the pipe when combined with some internal pressure, p_w
 F = safe working two-edge crush strength of pipe (kN/m)
 Y = factor of safety against crushing when an external load, W_c , is applied together with an internal pressure
 P_1 = internal pressure (kN/m^2) which will fracture the pipe when acting in combination with some external load, F_1 , applied in two-edge bearing
 P_2 = internal hydrostatic pressure (kN/m^2) which will burst the pipe in the absence of any external load.

External loading

From preceding considerations it is evident that the safe working two-edge crushing strength, F , corresponding to a particular maximum sustained operating (or static) pressure, p_w , must be compared with the total external load acting on the pipe due to backfill and any other superimposed loading.

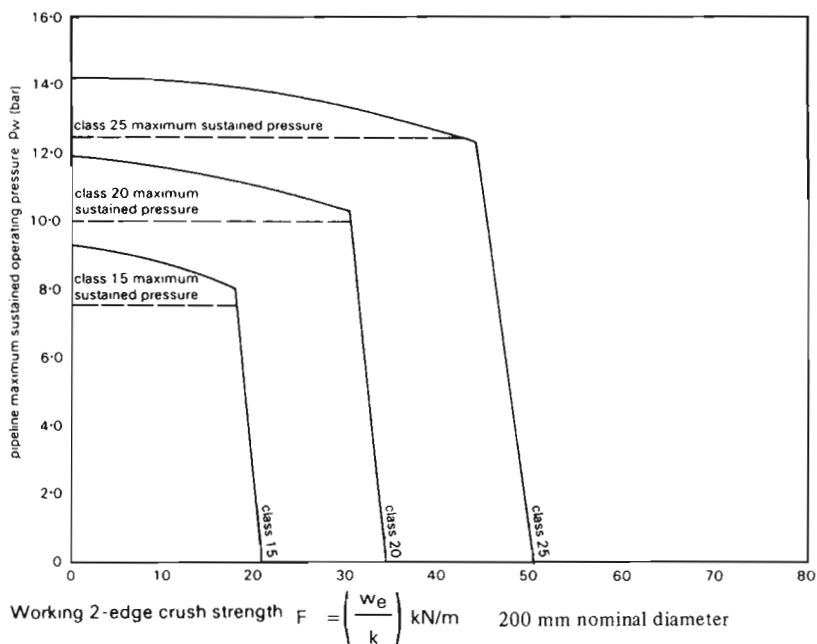


Chart 1. Example of combined loading chart: 200 mm diameter.

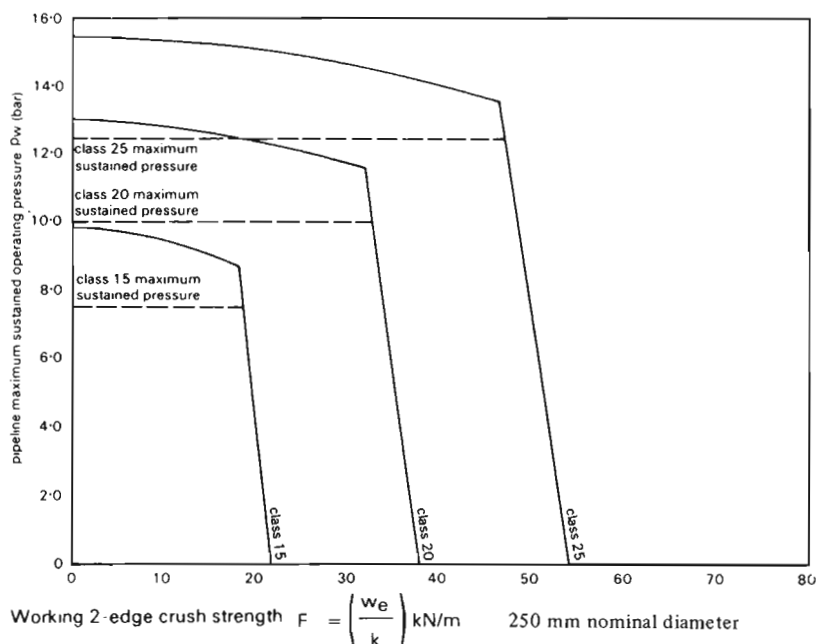


Chart 2. Example of combined loading chart: 250 mm diameter.

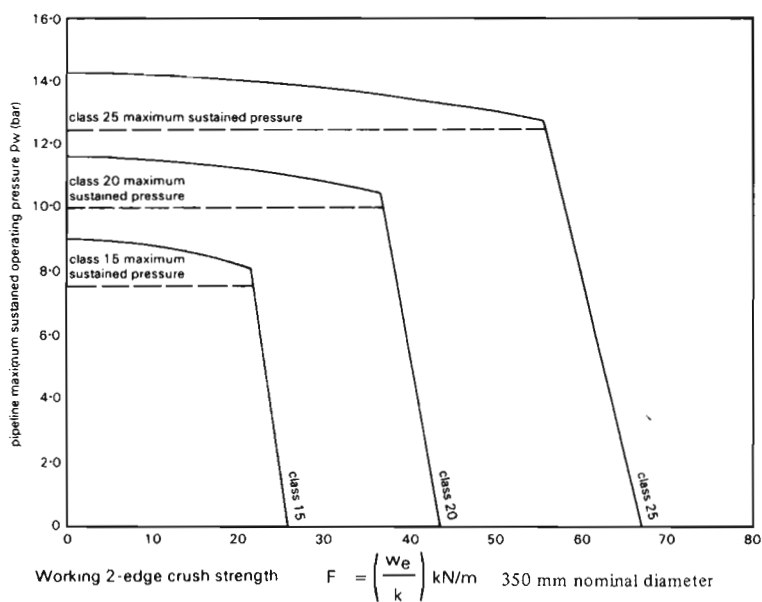


Chart 3. Example of combined loading chart: 350 mm diameter.

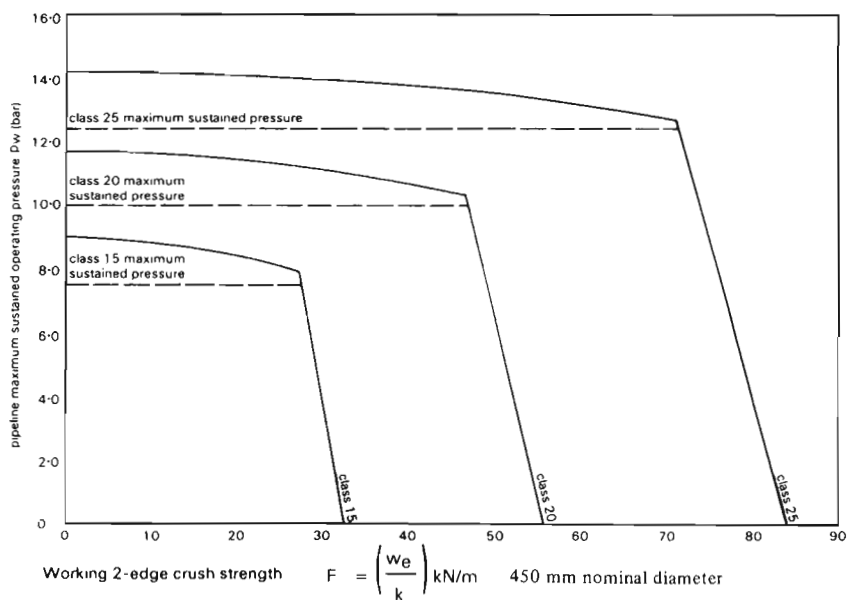


Chart 4. Example of combined loading chart: 450 mm diameter.

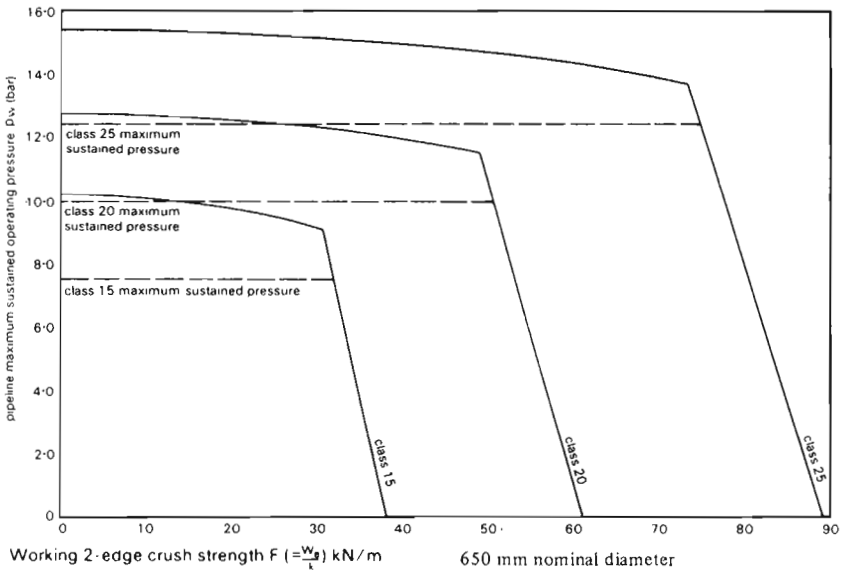


Chart 5. Example of combined loading chart: 650 mm diameter.

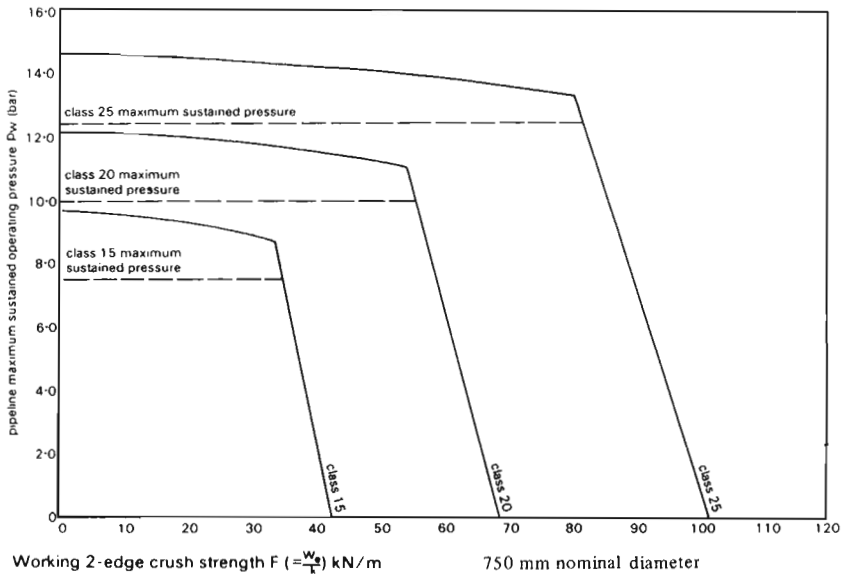


Chart 6. Example of combined loading chart: 750 mm diameter.

Table 1. Factors of safety (combined loading)

Nominal diameter of pipe			
mm	in	X	Y
200 and 225	7.9 and 8.9	3.5	2.5
250–500	9.8–19.7	3.0	2.5
600–750	23.6–29.5	2.5	2.5

Table 2. Bedding factors—Class 'C' bedding

Bedding angle α	Bedding factors (k) in different laying and backfill conditions		
	Trench and negative projection		Positive projection
	Ordinary backfill compacted between XX and YY	Ordinary backfill non-compacted between XX and YY	Ordinary compaction
30°	1.3	1.1	1.4
60°	1.5	1.2	1.7
90°	1.7	1.3	1.9
120°	1.7	1.3	1.9

Table 3. Bedding factors—Class 'B' bedding

Bedding angle α	Bedding factors (k) in different laying and backfill conditions		
	Trench and negative projection		Positive projection
	High compaction*	Ordinary compaction	Ordinary compaction
90°	2.6	1.9	2.3
120°	3.0	2.2	2.5

*At least 90% of maximum possible at the optimum moisture content (90% standard Proctor).

Notes:

1. The above factors of safety include allowance for surge provided that the maximum sustained operating, or static, pressure plus surge (i.e. pipeline design pressure) does not exceed the maximum allowable sustained pressure for the class of pipe by more than 10%.
2. For pipe sizes up to 150 mm (6 in) diameter, combined loading may not need to be considered as, in this range, pressure pipes are designed on the basis of beam strength and consequently have bursting and crushing strengths in excess of practical needs.

* F is defined in terms of a two-edge strength, so the following relationship is applicable:

$$kF = W_e \quad \text{or} \quad F = \frac{W_e}{k}$$

where W_e = total external loading acting on the buried pipe (kN/m)

k = bedding factor (see Tables 2 and 3).

The magnitude of W_e can be calculated from the appropriate Marston formulae and coefficient according to the type of soil and whether laid in trench ('narrow' or 'wide') or embankment conditions.

Table 4A. Modulus of soil reaction for pipes to EN 545

DN	80–300	350–450	500–1000
K (2 α)	0.110 (20°)	0.105 (45°)	0.103 (60°)
$\beta=0.75$ $E' = 0$	0.3–10.5	0.3–7.5	0.5–2.0
$E' = 1000$	0.3–11.0	0.3–8.5	0.3–3.5
$E' = 2000$	0.3–12.0	0.3–9.0	0.3–4.5
$E' = 5000$	0.3–14.0	0.3–12.0	0.3–8.5
$\beta=1.50$ $E' = 0$	0.3–10.5	0.4–7.0	See note 1
$E' = 1000$	0.3–11.0	0.4–8.0	0.6–3.0
$E' = 2000$	0.3–11.5	0.3–9.0	0.5–4.5
$E' = 5000$	0.3–14.0	0.3–12.0	0.3–8.0

Note 1: Only a specific calculation for each case can provide an adequate answer.

Note 2: The values given for the heights of cover have been established for the class K9; they are also valid for classes K $\geq$ 10.

Table 4B. Modulus of soil reaction for pipes to EN 598

DN	100–300	350–450	500–1000
K (2 α)	0.110 (20°)	0.105 (45°)	0.103 (60°)
$\beta=0.75$ $E' = 0$	0.3–5.0	0.5–3.0	0.5–2.0
$E' = 1000$	0.3–6.0	0.4–4.0	0.4–3.5
$E' = 2000$	0.3–6.5	0.3–5.0	0.3–4.5
$E' = 5000$	0.3–8.5	0.3–8.05	0.3–8.0
$\beta=1.5$ $E' = 0$	0.6–5.0	See note	See note
$E' = 1000$	0.5–5.5	0.7–3.51	0.8–3.0
$E' = 2000$	0.5–6.5	0.6–5.0	0.6–4.5
$E' = 5000$	0.4–8.5	0.4–8.0	0.4–7.5

Note: Only a specific calculation for each case can provide an adequate answer.

Trench beddings

Wherever soil and other related conditions permit, it has been widely adopted practice over many years to lay concrete pressure pipes on the well-levelled and prepared natural bottom of the trench. Selected backfill is introduced in layers not exceeding 300 mm (12 in), and properly compacted up to a level of 300 mm (12 in), approximately above the crown of the pipe. In the International Standard, beddings of this type are designated Class 'C'. This bedding embraces the Class 'C' and Class 'D' beddings described in National Building Studies Special Report No. 35. Consult EN545: 1994 and EN598: 1995. See also the chapter on Buried Pipes.

Figure 1. Layout diagram of a boiler-feed pump.

To meet potential requirements in individual projects regarding small positive suction lifts, a so-called 'booster pump' is installed before the boiler-feed pump. The positive suction lift of the booster pump is determined in the same way as for the boiler-feed pump.

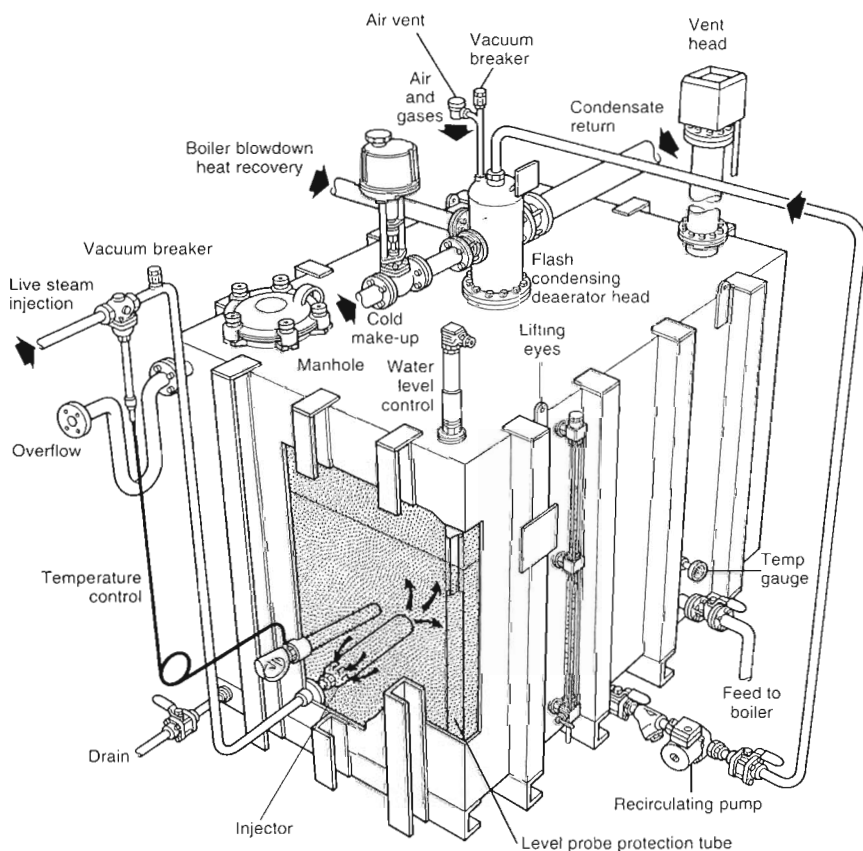
Minimum pressure in the suction branch (p_s) of the boiler-feed pump and/or the booster pump is given by the relation:

$$p_s = p_p + \frac{(h - H_{zs})\gamma}{10} \text{ (bar)}$$

where p_p = vapour tension pressure at a given temperature (bar)

γ = specific weight of water at a given temperature (kg/dm^3)

$$p_s = p_p + \frac{(h - H_{zs})\gamma}{148} \text{ (lbf/in}^2\text{)}$$



Feed tank design.

- where h and H_{zs} are in feet
 γ is in lb/ft^3

Pressure in the discharge branch (p_v) is given by:

$$p_v = p_k + \frac{h_{zv} \times \gamma}{10} \text{ (bar)}$$

where p_k = steam generator (boiler) pressure (bar)

h_{zv} = pressure losses in delivery piping from the boiler-feed pump branch as far as the boiler (m wg)

$$p_v = p_k + \frac{h_{zv} \times \gamma}{148} \text{ (lbf/in}^2\text{)}$$

where h_{zv} is in feet

γ is in lb/ft^3

Boiler-feed pump head

The head to be generated by the boiler-feed pump should be calculated from the required pressure in the discharge branch of the boiler-feed pump using the following relation:

$$H = \frac{p_v - p_s}{\gamma} \times 10 \text{ (m wg)}$$

$$H = \frac{p_v - p_s}{\gamma} \times 1755 \text{ (in wg)}$$

where pressures are in lbf/in^2

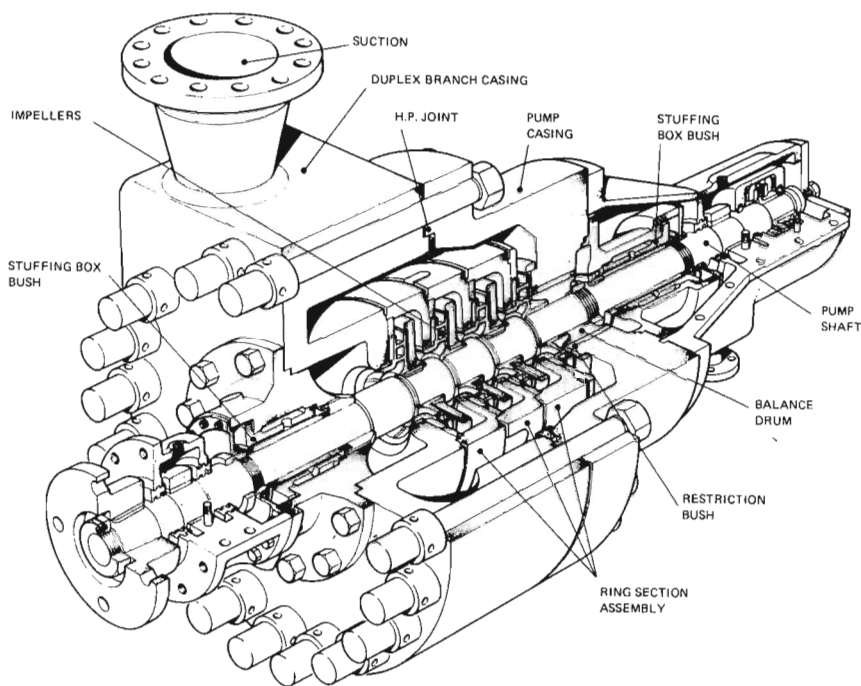
γ in lb/ft^3

However, the project engineer should take into account that the pump head in (m wg) will change its characteristics consistently, regardless of the feed-water temperature.

The specific weight of the water will change with its temperature. If a booster pump is to be installed before the boiler-feed pump, the same relation should be applied for calculating the pump head as that used for the calculation of the pump head when no booster is involved, so that:

$$H = \frac{p_v - p_s}{\gamma} \times 10 \text{ (m wg)}$$

$$H = \frac{p_v - p_s}{\gamma} \times 1755 \text{ (in wg)}$$



Compensated boiler-feed pump.

The only difference will be in establishing the magnitude of pressure in the inlet branch of the boiler-feed pump which can be calculated by the relation:

$$p_s = p_{vn} - \frac{h_{zp} \times \gamma}{10} \text{ (bar)}$$

where p_{vn} = discharge branch pressure of booster pump (bar)

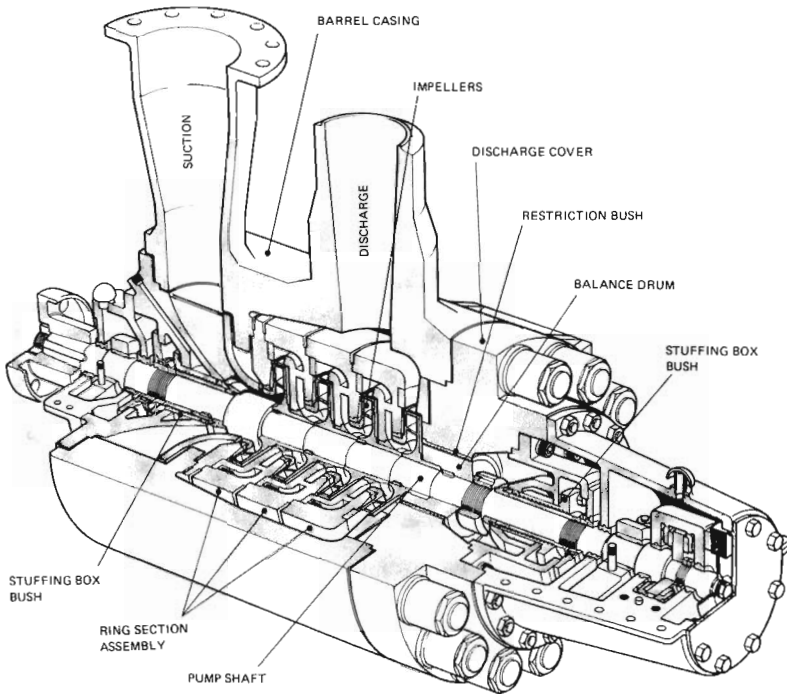
h_{zp} = head losses in the piping between the booster pump and boiler-feed pump (m)

$$p_s = p_{vn} - \frac{h_{zp} \times \gamma}{148} \text{ (lbf/in}^2\text{)}$$

for h_{zp} in feet
 γ in lb/ft³

Pump delivery

The boiler-feed pump delivery may be expressed as (l/hr) in project documents. The pumps are, however, tested in the manufacturer's test shops with cold water. In European practice, pump delivery is given in (l/min) and/or (l/sec).



Barrel-casing boiler-feed pump.

The relation between these deliveries will be:

$$Q_{(l/hr)} = Q_{(l/min)} \times \frac{\gamma \times 60}{1000}$$

Similarly, for Q in gal/min:

$$Q_{(gal/hr)} = Q_{(gal/min)} \times 60$$

Shaft input power required is given by:

$$N = \frac{Q \times (p_v - p_s)}{\gamma \times \eta} \times 0.02723 \text{ (kW)}$$

where Q = pump delivery in (l/hr)

p_v = discharge branch pressure (bar)

p_s = inlet branch pressure (bar)

γ = feed water specific weight (kg/dm^3)

η = boiler-feed pump efficiency at a calculated point (%)

In English units:

$$\begin{aligned}
 N &= \frac{Q_1 \times (p_v - p_s)}{\gamma \times \eta} \times 0.009325 \text{ (kW)} \\
 &= \frac{Q_1 \times (p_v - p_s)}{\gamma \times \eta} \times 0.0125 \text{ (hp)} \\
 &= \frac{Q_2 \times (p_v - p_s)}{\gamma \times \eta} \times 0.00777 \text{ (kW)} \\
 &= \frac{Q_2 \times (p_v - p_s)}{\gamma \times \eta} \times 0.010 \text{ (hp)}
 \end{aligned}$$

where Q_1 is in UK gal/min
 Q_2 is in US gal/min
 p_v and p_s are in lbf/in²
 γ is in lb/ft³.

The pump shaft input of a boiler-feed pump should be given by the manufacturer in the consistent units given later and calculated by the following equation:

$$N = \frac{Q \times H \times \gamma}{102 \times 60 \times \eta} = \frac{N_u}{\eta} \text{ (kW)}$$

where Q = pump delivery (l/min)
 H = head (m wg)
 N_u = pump useful capacity (kW)
 γ = feed-water specific gravity (kg/m³)

In English units (head H is in wg):

$$\begin{aligned}
 N &= \frac{Q_1 \times H \times \gamma}{51,500 \times \eta} \text{ (kW)} \\
 &= \frac{Q_1 \times H \times \gamma}{38,420 \times \eta} \text{ (hp)} \\
 N &= \frac{Q_2 \times H \times \gamma}{42,920 \times \eta} \text{ (kW)} \\
 &= \frac{Q_2 \times H \times \gamma}{3200 \times \eta} \text{ (hp)}
 \end{aligned}$$

For the selection of the driving machine, a planning margin in its output should be considered owing to the inaccuracies in calculations of the whole system and unpredictable conditions. This is why the driving machine output (N_M) will be obtained by the following relation:

$$N_M = (1.08 - 1.2)N$$

When speed-increasing gear boxes (speed-reducing gear boxes for booster pumps) and hydraulic couplings are to be set, then efficiency should also be taken into account for calculating the pump shaft input so that:

$$N = \frac{Q \times H \times \gamma}{102 \times 60 \times \eta \times \eta_p \times \eta_{sp}}$$

where η_p = gear box efficiency
 η_{sp} = hydraulic coupling efficiency

If the pump was tested with a cold-water supply then the following recalculations of the pump efficiency should be carried out if pumping hot water.

$$\eta_p = \eta_{ip} = \eta_m$$

where $\eta_{ip} = 1 - (1 - \eta_i) \times \frac{\gamma_p}{\gamma} \times 0.1$

$$\eta_i = \frac{\eta}{\eta_m} - \text{efficiency after subtracting bearing losses}$$

η_m = mechanical efficiency

γ = kinematic viscosity of water at the test shop

γ_p = kinematic viscosity of warm water

Leak-off

The whole of the pump shaft input, N , is not utilised in the pump for increasing the energy of the liquid, as part of the input that corresponds to all losses within the pump will be converted into heat. This means that the water temperature in the boiler-feed pump increases proportionally to the losses within the pump.

The temperature will then be:

$$\Delta t = \frac{860 N(1 - \eta)}{1000(Q + Q_k)} (^{\circ}\text{C})$$

where Q = delivery (l/hr)

Q_k = amount of water flow through the balancing device (l/hr)

N = pump shaft input (kW)

If the amount of the feed-water flow through the balancing device is introduced before the boiler-feed pump, the temperature of water entering the boiler-feed pump increases to the value:

$$t_1 = \frac{(Q_{t_0} + Q_k)t_k}{Q + Q_k}$$

where t_0 = water temperature in the inlet branch less the effect of water from the balancing device ($^{\circ}\text{C}$)

t_k = water temperature after the balancing device ($^{\circ}\text{C}$).

A temperature rise of approximately 10°C is permissible in small- and medium-sized boiler-feed pumps. At this point, a device discharging this leak-off delivery (from the piping placed after the boiler-feed pump) should come into operation automatically. In high-pressure boiler-feed pumps, more detailed analysis should be made of the leak-off amount (i.e. minimum delivery) at which the leak-off device must start to open automatically.

Steam Flow Calculations

Water freezes at 0°C (32°F) and boils at 100°C (212°F) under normal atmospheric pressure. 0°C (32°F) is the reference phase for zero heat content. Because the actual boiling point is dependent on ambient pressure, this can be designated t_1 . The heat required to raise the temperature of water from 0°C (32°F) to t_1 , marking the onset of vaporisation, is known as the sensible heat (h). At atmospheric pressure, the sensible heat of water is 349.3 kJ/kg (150.17 Btu/lb).

If heat continues to be applied to the water, the process of vaporisation (boiling) goes on until all the water has been transformed to steam (AB in Figure 1). During this period the temperature remains constant. The heat absorbed is called the latent heat of vaporisation (L), so that at point B:

$$\text{Total heat absorbed (H)} = h + L$$

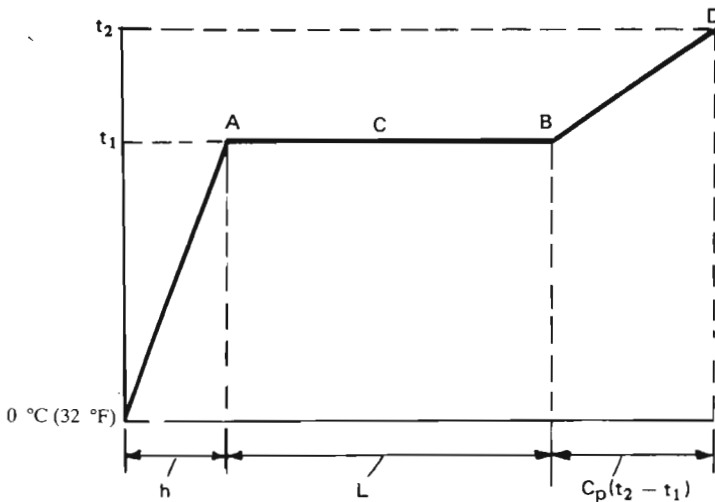


Figure 1.

Again, at atmospheric pressure, $L = 2259.7 \text{ kJ/kg}$ (970.3 Btu/lb) so that:

$$H = 349.3 + 2259.7$$

$$H = 2609 \text{ kJ/kg}$$

or

$$H = 150.17 + 970.3$$

$$H = 1150.5 \text{ Btu/lb}$$

At any intermediate point (C) between A and B, the stage of vaporisation can be expressed as the dryness fraction (q), or ratio of latent heat at that point to that necessary to produce a state of dry saturation (point B):

$$\text{Thus at any point C: } H = h + qL$$

If, after point B, more heat is added to the dry saturated steam, the steam is said to be superheated. Provided the steam is subject to unrestricted expansion, the pressure remains the same, but the temperature rises to t_2 . The degree of superheat is then $C_p(t_2 - t_1)$.

The approximate value of C_p is 0.48. Specific values are obtained from steam tables.

Steam can thus exist in three forms: dry saturated steam free of any water particles in suspension, wet saturated steam containing water particles in suspension, and superheated steam. The higher the degree of superheat the steam possesses, the more closely its characteristics resemble those of a perfect gas.

Steam flow through pipes

The original (Babcock) formula for determining the quantity of steam able to pass through a particular pipe with a specified pressure drop is:

$$W = 87.5 \sqrt{\frac{D(P_1 - P_2)d^5}{L(1 + 3.6/d)}}$$

where W = quantity of steam (lb/min)

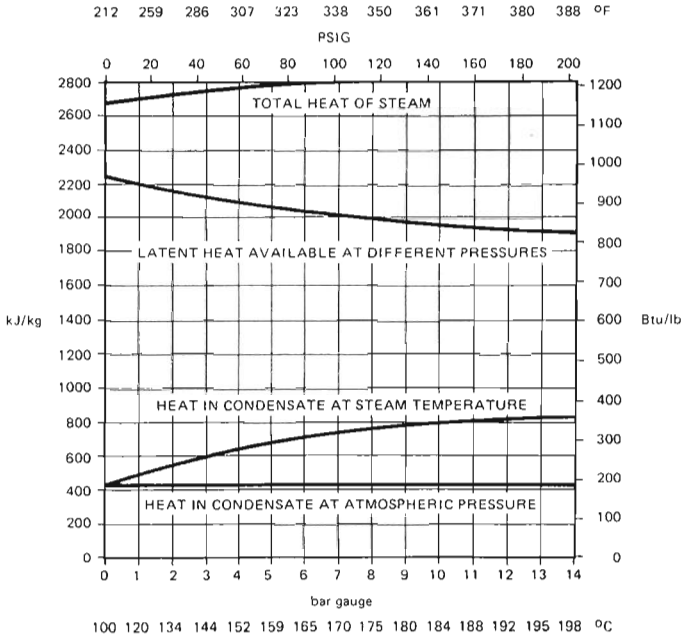
D = density of steam (lb/ft³)

P_1 = initial pressure of steam (lbf/in²)

P_2 = final pressure of steam at end of pipe (lbf/in²)

d = inside diameter of pipe (in)

L = length of pipe (ft)



Changes in amounts of heat required for the two stages of steam production.

The formula can also be rewritten as a solution for pressure drop:

$$(P_1 - P_2) = 0.0001306 \frac{W^2 L (1 + 3.6/d)}{D \times d^5}$$

These formulae are applicable to both saturated steam and superheated steam.

A further formula derived from these gives the size of pipe required for a specific mean velocity of steam:

$$d = 1.75 \sqrt{\frac{W}{V \times D}}$$

where V = steam velocity (ft/sec)

Typical values of V are:

- 20 to 30 m/s (70–100 ft/sec) for exhaust steam
- 30 to 45 m/s (100–150 ft/sec) for saturated steam
- 40 to 60 m/s (130–200 ft/sec) for superheated steam.

Such formulae may still be used for simplified (approximate) calculations, but modifications of the D'Arcy equation are now normally employed, viz:

$$\begin{aligned}(P_1 - P_2) = \Delta P &= W^2 \left(\frac{0.000336f}{d^5} \right) \times \bar{V} \\ &= W^2 \left(\frac{33,600f}{d^5} \right) \times \bar{V} \times 10^{-9}\end{aligned}$$

where W is the flow rate (lb/hr)

$\bar{V}$ is the specific volume (ft³/lb)

f is the friction factor appropriate to the pipe and flow velocity.

Working is further simplified by calculating a discharge factor (C_1):

$$C_1 = W^2 \times 10^{-9}$$

and a size factor (C_2):

$$C_2 = \frac{33,600f}{d^5}$$

(Tabular data are available for determining C_2 .)

Then:

$$\begin{aligned}\Delta P &= C_1 C_2 \bar{V} \\ &= \frac{C_1 C_2}{\rho}\end{aligned}$$

where ρ is the mass density of the steam (lb/ft³) and is dependent on temperature and pressure.

$$W = \sqrt{C_1 \times 10^{-9}}$$

$$d = \sqrt[5]{\frac{33,600f}{C_2}}$$

$$Q = \frac{W}{4.8}$$

where Q = flow rate (ft³/min at STP)

Sizing of condensate-return lines

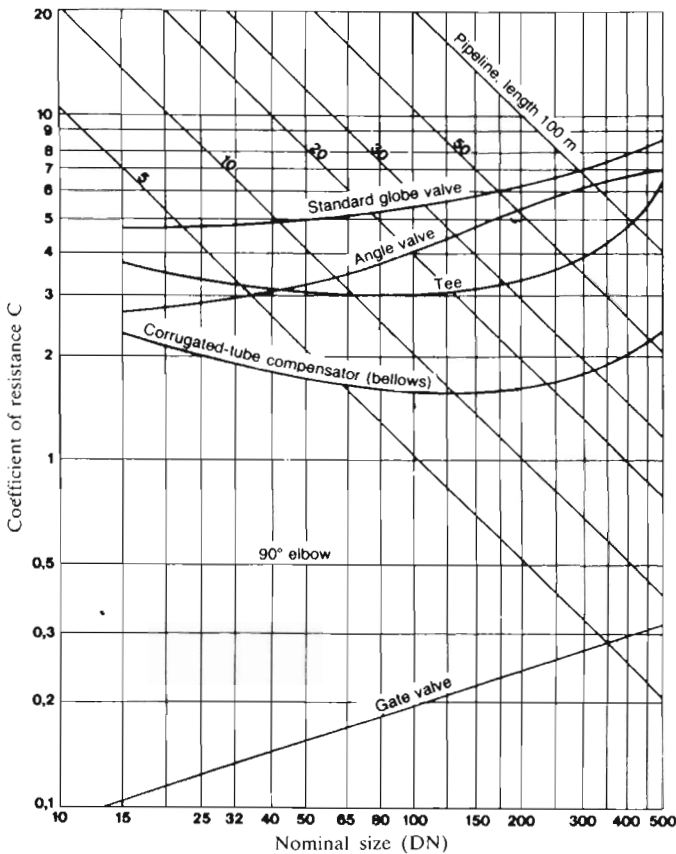
Basic considerations

The diameter of the pipeline between the heat exchanger and steam trap is normally chosen to fit the nominal size of the trap.

When choosing the diameter of the condensate line downstream of the trap, flashing has to be considered. Even at very low pressures the volume of flash steam is many times that of the liquid if the condensate is at saturation temperature upstream of the trap (e.g. during flashing from 1.2 to 1 bar (17 to 14.4 lbf/in²) the volume increases approximately 17 times).

In these cases it is possible to dimension the condensate line in accordance with the amount of flash steam formed. The flow velocity of the flash steam should not be too high otherwise water hammer (by the formation of waves), flow noises and erosion may occur.

A flow velocity of 15 m/s (50 ft/sec) at the end of the pipeline before the inlet into the collecting tank or flash vessel is a useful empirical value. The inside diameter of the pipeline required can be taken from Table 1.



The coefficients of resistance C of all pipeline components of the same size are read in the above chart. The total pressure drop Δp in bar can be determined with the sum of all individual components (ΣC) and the operating data, (see Figure 5).

Figure 2. Pressure drops in steam lines.

Table 1. Condensate line sizing (based on flash steam)

State of condensate before flashing		Pressure at end of condensate line (bar a)																					
Pressure bar a	Related boiling temperature °C	0.2	0.5	0.8	1.0	1.2	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	6	7	8	9	10	12	15	18	20
1.0	99	35.7	16.0	7.4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1.2	104	37.9	18.0	10.0	6.1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1.5	111	40.1	20.6	12.9	9.5	6.8	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2.0	120	44.2	23.5	15.8	12.6	10.3	7.6	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2.5	127	46.8	25.5	17.7	14.5	12.3	9.2	5.3	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
3.0	133	48.8	27.1	19.2	16.0	13.9	10.7	7.3	4.5	—	—	—	—	—	—	—	—	—	—	—	—	—	—
3.5	138	50.4	28.4	20.4	17.1	15.0	11.9	8.5	6.0	3.8	—	—	—	—	—	—	—	—	—	—	—	—	—
4.0	143	52.0	29.6	21.5	18.2	16.0	12.9	9.7	7.3	5.3	3.5	—	—	—	—	—	—	—	—	—	—	—	—
4.5	147	53.3	30.5	22.3	19.0	16.9	13.7	10.5	8.1	6.3	4.7	3.0	—	—	—	—	—	—	—	—	—	—	—
5	151	54.3	31.5	23.1	19.8	17.7	14.4	11.2	8.9	7.1	5.6	4.2	2.8	—	—	—	—	—	—	—	—	—	—
6	155	55.7	32.3	23.9	20.5	18.4	15.2	11.9	9.6	7.9	6.5	5.1	4.0	2.7	—	—	—	—	—	—	—	—	—
7	158	56.5	33.0	24.5	21.1	18.9	15.7	12.4	10.1	8.4	7.0	5.7	4.6	3.5	2.1	—	—	—	—	—	—	—	—
8	170	59.9	35.5	26.7	23.1	20.9	17.6	14.2	11.9	10.2	8.9	7.7	6.7	5.8	4.8	4.0	—	—	—	—	—	—	—
9	175	61.3	36.4	27.5	23.9	21.7	18.3	14.9	12.6	10.9	9.5	8.4	7.4	6.6	5.5	4.8	2.4	—	—	—	—	—	—
10	179	62.3	37.2	28.2	24.6	22.3	18.9	15.5	13.1	11.4	10.0	8.9	7.9	7.1	6.0	5.3	3.3	2.1	—	—	—	—	—
12	187	64.4	38.7	29.5	25.7	23.5	19.9	16.5	14.1	12.3	11.0	9.8	8.9	8.0	7.0	6.2	4.5	3.6	2.8	—	—	—	—
15	197	66.9	40.5	31.0	27.2	24.8	21.5	17.7	15.2	13.4	12.0	10.8	9.9	9.1	8.0	7.2	5.6	4.8	4.2	2.9	—	—	—
18	206	69.0	42.0	32.3	28.4	26.0	22.3	18.7	16.2	14.3	12.9	11.7	10.8	9.9	8.8	8.0	6.5	5.7	5.1	3.9	2.5	—	—
20	211	70.2	42.9	33.0	29.0	26.6	22.9	19.2	16.7	14.8	13.4	12.2	11.2	10.4	9.2	8.4	7.0	6.2	5.6	4.4	3.1	1.7	—
25	223	72.9	44.8	34.7	30.6	28.1	24.2	20.4	17.9	15.9	14.5	13.2	12.2	11.4	10.2	9.3	7.9	7.1	6.5	5.4	4.2	3.1	2.5
30	233	75.1	46.3	36.0	31.8	29.2	25.3	21.4	18.8	16.8	15.3	14.0	13.0	12.1	10.9	10.0	8.6	7.8	7.2	6.1	4.9	4.0	3.4
35	241	76.8	47.5	37.0	32.7	30.1	26.1	22.1	19.5	17.5	15.9	14.6	13.6	12.7	11.4	10.5	9.2	8.4	7.8	6.7	5.5	4.5	4.0
40	249	78.5	48.7	38.0	33.6	31.0	26.9	22.9	20.1	18.1	16.5	15.2	14.1	13.2	12.0	11.0	9.7	8.6	8.2	7.1	6.0	5.0	4.5
45	256	80.0	49.7	38.8	34.4	31.7	27.5	23.5	20.7	18.6	17.0	15.7	14.6	13.7	12.4	11.4	10.1	9.3	8.6	7.5	6.3	5.4	4.9
50	263	81.4	50.7	39.6	35.2	32.5	28.2	24.1	21.2	19.1	17.5	16.2	15.1	14.2	12.8	11.8	10.5	9.6	9.0	7.9	6.7	5.7	5.2

To determine the actual diameter, the above values must be multiplied with the following factors:

kg/h	100	200	300	400	500	600	700	800	900	1000	1500	2000	3000	5000	8000	10,000	15,000	20,000
Factor	1.0	1.4	1.7	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.9	4.5	5.5	7.1	8.9	10.0	12.2	14.1

Bases for determining the inside diameter:

- 1. The flash steam amount only is being considered.
- 2. The flow velocity of the flash steam is assumed to be 15 m/s.

For long pipelines (over 100 m or 300 ft) and large condensate flowrates, the pressure drop should be calculated to avoid the back pressure becoming too high. The velocity of the flash steam may be used in the calculations (see Table 1 and Figure 2).

When the condensate is mainly in the liquid state (e.g. high degree of undercooling, extremely low pressure) its flow velocity should, if possible, be rated at 0.5 m/s (1.64 ft/sec) or higher. The pipeline diameter can be chosen from the chart (Figure 3).

If the condensate is pumped, the condensate in the pump discharge line can only be in the liquid phase. For choosing the pipeline diameter, the mean velocity can be rated at 1.5 m/s (5 ft/sec). Again, Figure 3 may be used.

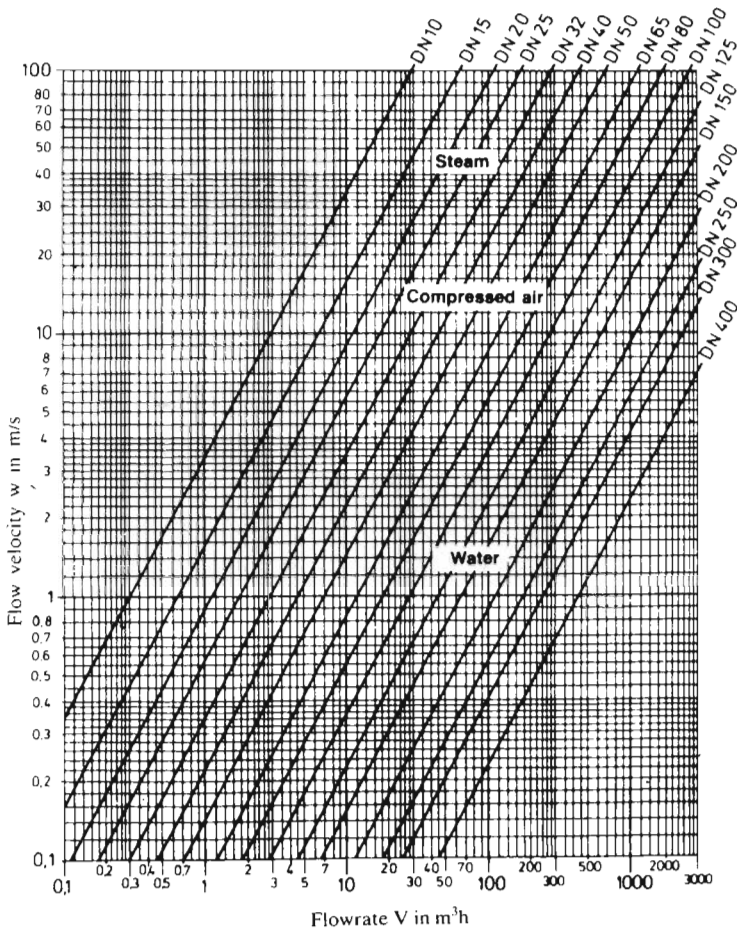


Figure 3. Volume flowrates in pipelines.

Calculation of condensate flowrates*Basic formula*

If the amount of heat required in kcal/hr is known (indicated on the name plate of the heat exchanger) or easy to determine, the condensate flowrate M can be calculated:

$$M = 1.2 \frac{\text{kcal/hr}}{500} \text{ (kg/hr)}$$

The quotient 500 is the latent heat of steam (kcal/kg) for medium pressures. The factor 1.2 is added to compensate for the heat losses.

In SI units, the condensate flowrate is calculated as follows:

$$M = 1.2 \frac{W}{2000} \times \frac{3600}{1000}$$

Hence:

$$M \approx 1.2 \frac{W}{560} \text{ (kg/hr)}$$

W is the amount of heat required in Watts or Joule per second (J/sec) and the quotient 2000 the latent heat of steam (kJ/kg) for medium pressures.

If the amount of heat Q per hour is not known, it can be calculated from the weight M of the product to be heated in 1 hour, the specific heat $\left(c = \frac{\text{kcal}}{\text{kgK}} \text{ or, in SI units, } c = \frac{\text{J}}{\text{kgK}} \right)$ and the difference between initial temperature

t_1 and final temperature t_2 ($\Delta t = t_2 - t_1$):

$$Q = M \times c \times \Delta t \text{ (kcal/hr)}$$

or in SI:

$$Q = M \times \frac{c}{3600} \times \Delta t \text{ (W)}$$

Example:

50 kg of water to be heated in 1 hour from 20 to 100°C. The amount of heat required is:

$$Q = 50 \times 1 \times (100 - 20) = 4000 \text{ kcal/hr}$$

or, in SI:

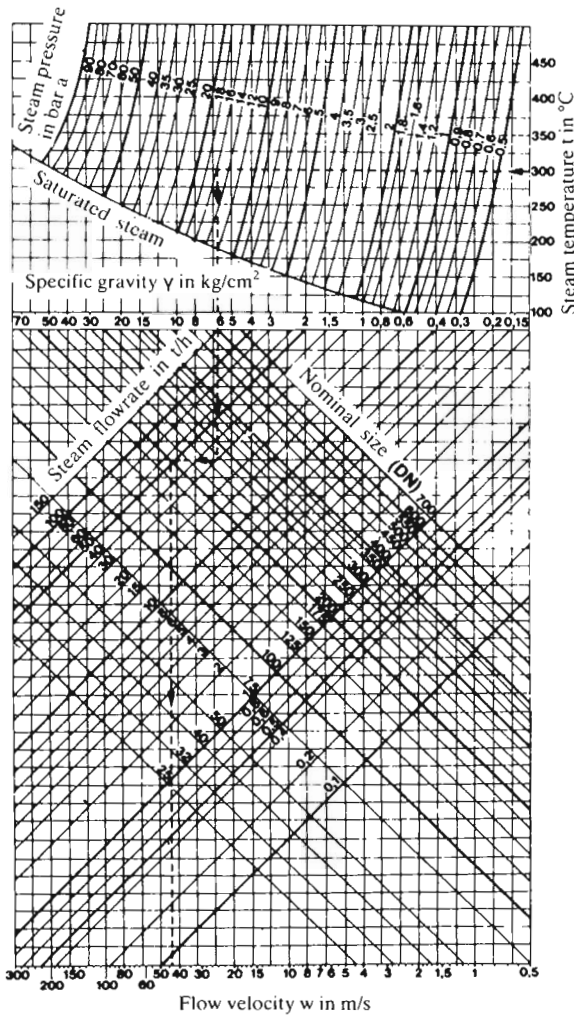
$$Q = 50 \times \frac{4187}{3600} \times (100 - 20) = 4652 \text{ W}$$

The amount of condensate is:

$$M = 1.2 \times \frac{4000}{500} = 9.6 \text{ kg/hr}$$

or, in SI:

$$M = 1.2 \times \frac{4652}{560} \times 9.97 \text{ kg/h}$$



Example:

Steam temperature 300 °C, steam pressure 16 bar, Result:
a steam flowrate 30 t/h, nominal size (DN) 200 mm. Flow velocity = 43 m/s.

Figure 4. Flow velocity in steam lines.

Example

Pipeline components DN 50 mm

Pipeline length 20 m	C = 8.1
1 angle valve	C = 3.1
1 standard globe valve	C = 5
1 tee	C = 3.1
3 elbows	C = 1.5

$$\sum C = 21.0$$

Operating data

Temperature	= 300°C
Steam pressure	= 16 bar a
Velocity	= 40 m/s

Result $\Delta p = 1.1$ bar

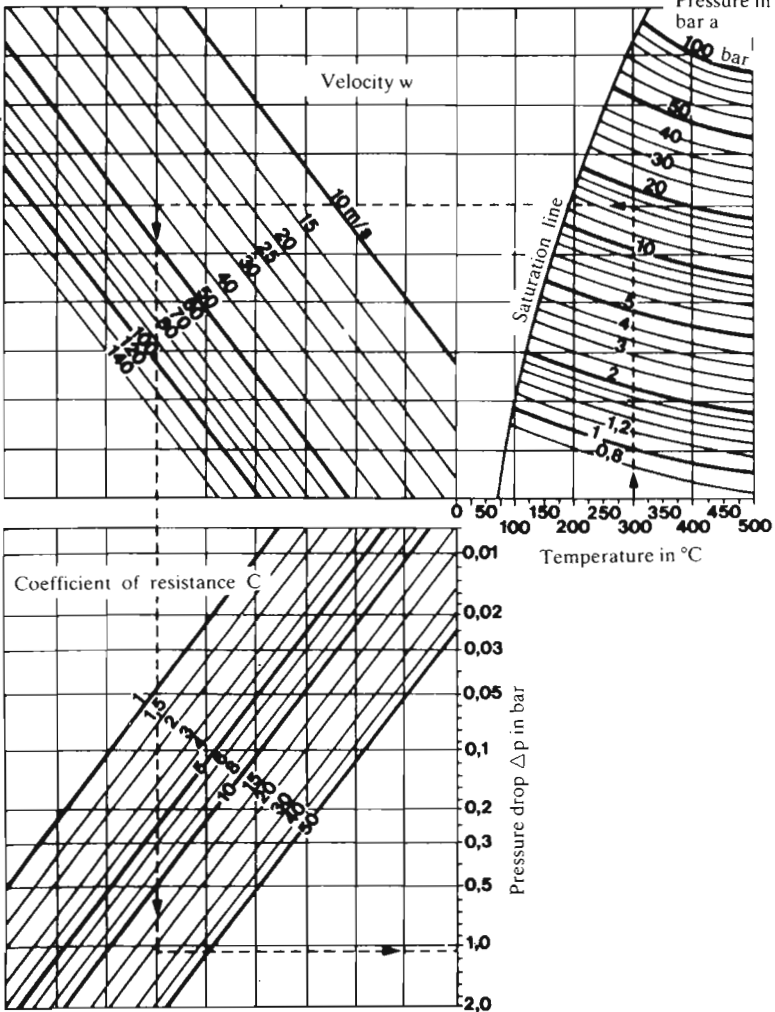


Figure 5. Pressure drops in steam lines.

Sizing of steam lines

When sizing steam lines, care must be taken that the pressure drop between the boiler and steam users is not too high. The pressure drop depends mainly on the flow velocity of the steam.

The following empirical values for the flow velocity have proven to be satisfactory:

Saturated steam lines 20–40 m/s (65–130 ft/sec)

Superheated steam lines 35–65 m/s (115–215 ft/sec)

The lower figures should be used for smaller flowrates.

For a given flow velocity, the required pipe diameter can be chosen from the chart (Figure 4).

The pressure drop can be calculated from the charts in Figures 2 and 5.

Calculation of condensate flowrates

If 50 kg of water is to be vaporised in 1 hour, the latent heat of approximately 5600 kcal/kg or 2000 kJ/kg has to be added:

$$50 \times 500 = 25,000 \text{ kcal/hr}$$

or, in SI:

$$50 \times 2000 = 100,000 \text{ kJ/h} = \frac{100,000 \times 1000}{3600} = 27,778 \text{ W}$$

The total amount of heat required, and consequently the total amount of condensate formed, can be calculated as follows:

$$M = 1.2 \times \frac{4000 + 25,000}{500} = 69.6 \text{ kg/h}$$

or, in SI:

$$M = 1.2 \times \frac{4652 + 27,778}{2000} \times \frac{3600}{1000} = 70 \text{ kg/h}$$

Each product has its own specific heat.

Calculation of condensate flowrates

If the size of the heating surface and the temperature rise (between initial and final temperatures) of the product are known, the condensate flowrate M can be calculated with sufficient accuracy as follows:

$$M = \frac{A \times k \left(t_s - \frac{t_1 + t_2}{2} \right)}{r} \text{ (kg/hr)}$$

Table 2. Heat emission from pipes

Heat emission from bare horizontal pipes with ambient temperatures between 10 and 21°C and still air conditions.

Temperature difference: steam to air °C	Pipe size									
	15 mm	20 mm	25 mm	32 mm	40 mm	50 mm	65 mm	80 mm	100 mm	150 mm
	W/m									
56	54	65	79	103	108	132	155	188	233	324
67	68	82	100	122	136	168	198	236	296	410
78	83	100	122	149	166	203	241	298	360	500
89	99	120	146	179	205	246	289	346	434	601
100	116	140	169	208	234	285	337	400	501	696
111	134	164	198	241	271	334	392	469	598	816
125	159	191	233	285	321	394	464	555	698	969
139	184	224	273	333	373	458	540	622	785	1060
153	210	255	312	382	429	528	623	747	939	1255
167	241	292	357	437	489	602	713	838	1093	1492
180	274	329	408	494	556	676	808	959	1190	1660

Heat emission from bare horizontal pipes with ambient temperatures between 50 and 70°F and still air conditions.

Temperature difference: steam to air °F	Pipe size									
	1/2 in	3/4 in	1 in	1 1/4 in	1 1/2 in	2 in	2 1/2 in	3 in	4 in	6 in
	Btu/linear ft/hr									
100	56	68	82	107	113	138	163	194	243	337
120	71	85	104	127	142	175	206	246	308	427
140	86	104	127	155	173	212	251	300	375	521
160	103	125	152	186	213	256	301	362	451	626
180	121	146	176	217	243	297	351	417	522	725
200	139	171	206	251	282	346	408	488	622	850
225	166	199	243	297	334	410	483	578	726	1009
250	192	233	284	347	389	478	563	674	849	1180
275	220	266	326	398	447	550	649	778	978	1360
300	251	304	372	455	510	628	742	888	1140	1557
325	285	343	425	520	580	705	843	1010	1240	1730

or, in SI:

$$M = \frac{A \times k \left(t_s - \frac{t_1 + t_2}{2} \right)}{r} \times \frac{3600}{1000} \text{ (kg/h)}$$

where M = amount of condensate (kg/h)

A = heating of surface (m^2)

k = coefficient of overall heat transfer ($\text{kcal/m}^2\text{h K}$)

or, in SI in $\frac{\text{W}}{\text{m}^2\text{K}}$

t_s = temperature of steam

t_1 = initial temperature of product

t_2 = final temperature of product (quite often it is sufficient if the average temperature is known, e.g. room temperature)

r = latent heat in kcal/kg or kJ/kg (approximation for medium pressures 500 kcal/kg or, in SI, 2000 kJ/kg)

A few empirical values for the coefficient of overall heat transfer k are given as follows:

	$\frac{\text{kcal}}{\text{m}^2\text{hr K}}$	$\frac{\text{W}}{\text{m}^2\text{K}}$
Insulated steam line	0.5–2	0.6–2.4
Non-insulated steam line	7–10	8–12
Unit heater with natural circulation	4–10	5–12
Unit heater with forced circulation	10–40	12–46
Jacketed boiling pan with agitator	400–1300	460–1500
As above, with boiling liquid	600–1500	700–1750
Boiling pan with agitator and heating coil	600–2100	700–2400
As above, with boiling liquid	1000–3000	1200–3500
Tubular heat exchanger	250–1000	300–1200
Evaporator	500–1500	580–1700
As above, with forced circulation	800–2600	900–3000

Cavitation

The phenomenon of cavitation is associated with a reduction of pressure occurring in a liquid system, reducing the liquid pressure at a localised area down to the vapour pressure of the liquid concerned. As a consequence, vapour and small gas-filled bubbles form in the liquid at this point and are entrained by the flow. As soon as they reach a region of higher pressure, they suddenly collapse at extremely high velocities, with vapour condensing into liquid again. Very high implosion pressures are generated, depending on the bubble size, and may even reach 10,000 bar (140,000–150,000 lbf/in²). Such high-velocity impacting pressures can show up as:

- (i) noise
- (ii) vibration (critical oscillations)
- (iii) mechanical damage to construction materials.

Critical regions for the development of cavitation in a piping system are sudden cross-section enlargements or contractions, changes in flow direction and sudden changes in flow (such as at throttling gaps). The sudden contraction, illustrated in Figure 1, is a typical example. This shows the pressure distribution at the throttling point, where p_1 is the upstream fluid pressure, p_2 the downstream fluid pressure (lower because of the head loss after throttling), p_A is the atmospheric pressure and p_0 the vapour pressure of the fluid. At point p_3 , the pressure is reduced to the vapour pressure, generating cavitation, with subsequent pressure recovery to p_2 . Somewhere between p_3 and p_2 the cavitation bubbles will collapse suddenly.

Cavitation does not necessarily lead to damage, even if it does generate noise and vibration. It depends on the intensity of cavitation or, specifically, the lifespan of the bubbles from formation to implosion. Cavitation intensity decreases with an increase in the life of the bubbles.

The pressure travel gradient (p_1 , p_3 , p_0) is thus significant and related to the shape of the flow passage. As a general rule, the pressure drop can be influenced by streamlining the flow, although this can be optimised for one predetermined flow condition only. Thus it is rather more important to try to extend the pressure-travel gradient. Geometric shapes can be found which, despite cavitation, do not lead to damage, i.e. the bubble implosion occurs

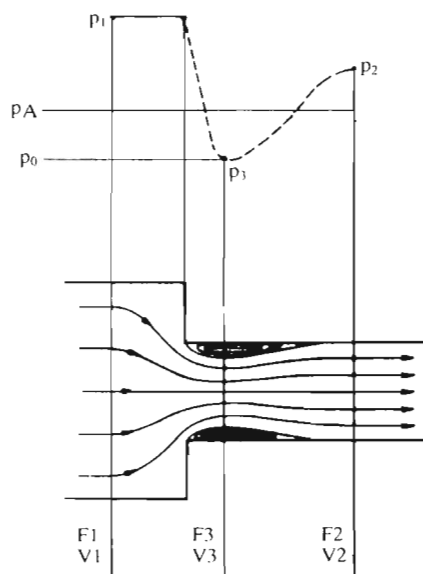


Figure 1.

away from the possibility of contact with the material surfaces. From this it can be seen that only bubble implosions near the wall are destructive. If the bubbles contact material surfaces, the destruction mechanism conforms with that of liquid droplet erosion.

From the point of view of metal physics, what happens is a high-velocity deformation of the metal as a result of the bubble implosion. In many cases the mechanical erosive influence is coupled with an electrochemical corrosive influence, cavitation and corrosion occurring together. It has been shown that, in the case of industrial water, damage to carbon steel and ingot iron can be reduced by cathodic protection, i.e. the corrosive influence can be removed. Among construction materials which have proved to be less prone to cavitation, austenitic steel, single-phase copper alloys (bronzes), stainless steels and stellite armouring have been most successful. These materials are largely resistant to corrosion so they are not subject to this additional attack.

Typical potential points of cavitation damage are:

- (a) suction pipes of pumps
- (b) narrow flow spaces, leakage and bearing gaps
- (c) sudden changes in flow area
- (d) changes in flow direction, as in bends and pipe tees
- (e) changes which lead to turbulence
- (f) downstream of throttle and control valves; components built into the flow stream.

From the following list, measures can be chosen by which disadvantageous cavitation consequences can be avoided:

- (1) Avoidance of turbulence by proper streamlining of flow, e.g. by means of a vaned ring in a needle valve (Figure 2).
- (2) Prevention of wall contact after regions of pressure drop by sudden enlargement of the pipeline (Figure 3). Developing cavitation bubbles implode in the water space. Cavitation arises in a space not endangering the material.
- (3) Letting pressure drop occur over sharp edges.
- (4) Dissipation of the kinetic energy not solely through turbulent mixing but:
 - (a) through built-in resistance, i.e. by increasing the friction-causing wall surfaces. This causes an increase in the back pressure after a point of throttling. A pressure drop below the atmospheric pressure can be avoided (Figure 4);
 - (b) partitioning of several resisting bodies in series (multi-stage pressure drop).

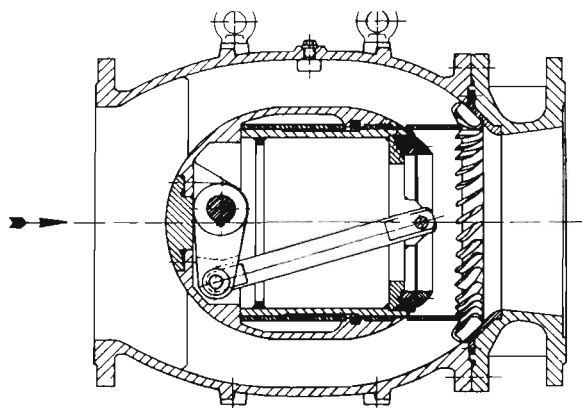


Figure 2.

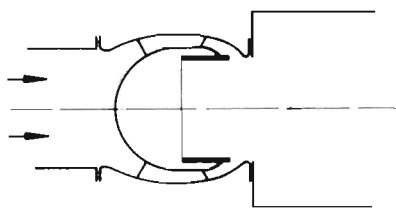


Figure 3.

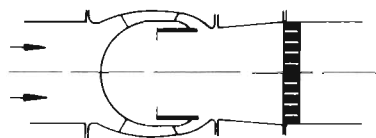


Figure 4.

As an approximation, the number of stages required can be calculated as follows:

$$n = -6.45 \times \lg \frac{p_2}{p_1}$$

where p_1 = upstream pressure
 p_2 = downstream pressure

- (5) Principle of flow partition into small single cross-sections, e.g. hollow cylinders (Figures 5 and 6). The division into single jets beneficially influences the dynamic behaviour of the medium flowing off downstream of the throttling devices. In addition, the partition into small cross-sections achieves a more uniform downstream flow in the following pipe.



Figure 5. Choke valve with low-recovery trim.

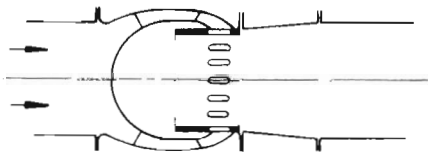


Figure 6.

(6) Change of flow contractions by the introduction of air:

- (a) in free exit, similar to the effect of perlators in water taps with the well-known soft jet (Figure 7);
- (b) intensive ventilation in the flow passage produces the hydraulic character of an end closure. The arrangement of the valve on the air side is, from the hydraulic point of view, more favourable than that within a pipeline with ventilation.

From the aforementioned it can be seen that, for throttling and control duties, special valve types are required which are designed with special seat and exit configurations. In general, valves are limited only according to the nominal pressure rating. Within the standard design requirements, the special dynamic loadings in the flow passage of the various types of valves are not considered. In the case of mere shut-off valves, such as gate and butterfly valves, the necessary adaptation cannot be achieved by means of design, (Figure 8). They are, therefore, not suitable for pronounced throttling and control duty but, because of the low permanent pressure drop, are ideal for on-off duty.

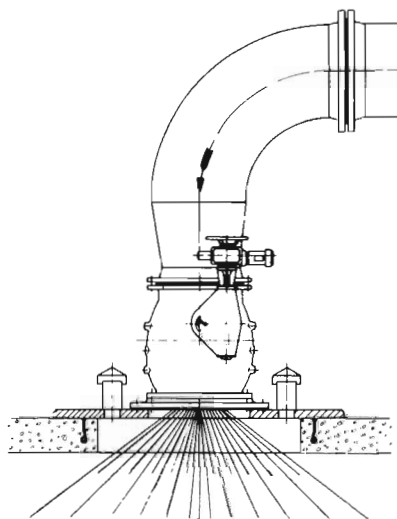


Figure 7.

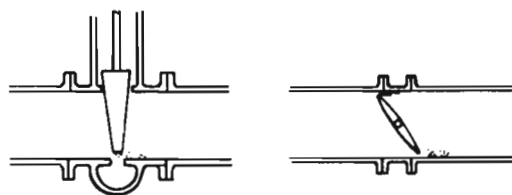


Figure 8.

Such valves are suitable for short-period throttling duties such as during shut-off in the case of a burst pipe. However, when dimensioning these valves, the limits which result from the energy head must be considered:

Butterfly valves:

PN25	7.5 m/s	(25 ft/sec)
PN16	5 m/s	(16.5 ft/sec)
PN10	4 m/s	(13 ft/sec)
PN4	2.5 m/s	(8 ft/sec)
PN2A5	2 m/s	(6.5 ft/sec)

(Flow velocities referred to valve nominal diameter.)

If butterfly valves are used as safety devices in the case of a burst pipe, the responsible manufacturer considers the stresses in these circumstances and dimensions valve and operator designs in a correspondingly strong manner.

Butterfly valves can be used for on-off operation, in conjunction with special parallel-gate valves with perforated fixed plates, for the continuous control of the downstream flow of water from dams.

The perforated plate divides the flow into a large number of small jets to create the required throttling effect. The jets are evenly distributed over the cross-section of the pipe and the uniform small-jet configuration achieved is effective in suppressing vibration, cavitation damage, pressure fluctuations and noise.

Components of the control, shown in Figure 9, are simply two circular perforated plates and an annular body (1) mounted between pipe flanges. Plate (2) is fixed, and plate (3), on the upstream side, is free to slide up and down. In the fully open position, the orifices in the plates coincide. The fully closed position is obtained by displacing the moving plate (3) through a distance equal to one orifice diameter. Under normal conditions of flow control, the position is intermediate with the orifices in the fixed plate only partially blocked off by those of the moving plate. The latter may be positioned by hand or by valve actuator.

In water works and water power plants, needle valves, also known as ring piston slide valves (Figure 10), have proved in more than 40 years of practice to be excellent as flow-control valves, because individual adaptation to the given operating conditions and duties is, with this type of valve, possible. With reference to cavitation, this means that, by specific configuration of the outlet shape, formation of the throttling point, design of the downstream piping, and by the selection of the point of installation, the hydraulic conditions can be influenced directly and damages due to cavitation be avoided. With these control valves, all intermediate positions, i.e. partial openings, must be possible for continuous duty to achieve variable flows or an effective change in energy, e.g. reduction of pressure. No ill effect due to cavitation or vibration

must occur. The design of the needle valves offers all the advantages. The flow is guided through a ring-shaped passage around a ball-shaped inside body.

The outside body is so designed that the free-flow cross-section continuously diminishes from the inlet to the sealing and throttling point so that flow velocity increases. Shortly before the narrowest cross-section, a vaned ring is provided, which swirls the outer flow filaments in such a way that the fluid is

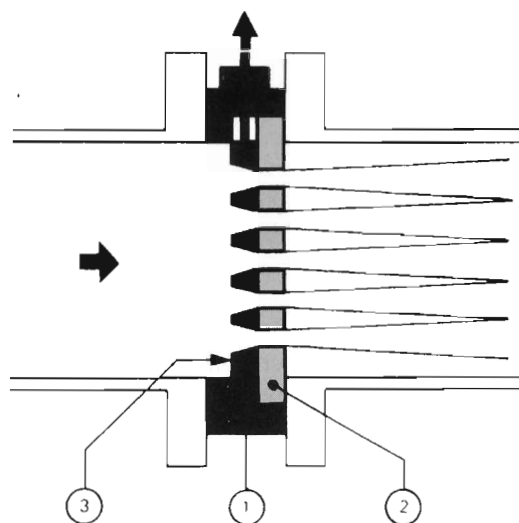


Figure 9. Schematic showing components of control for parallel-gate valve with perforated fixed plates.

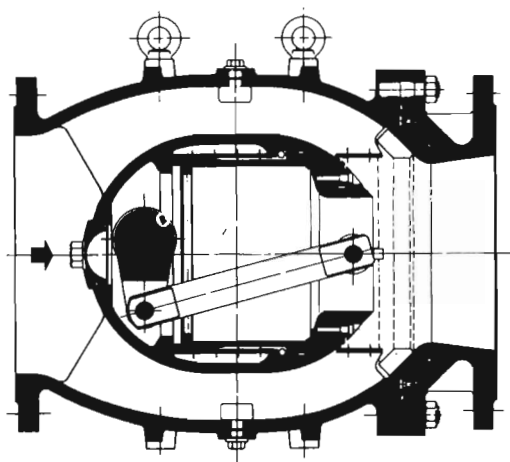


Figure 10. Typical ring-piston slide valve.

forced against the wall of the downstream flow section so that detachments are avoided and cavitation bubble implosions are kept away from the wall.

The shut-off piston in the spherical inside body moves in the opening and closing passage, i.e. in or against the direction of flow, and produces, therefore, a linear change in cross-section without causing the flow direction to change. The downstream shape of the piston is sharp-edged. In contrast to former designs with pointed piston ends, the hydraulic exit flow angle can develop,



A globe valve fitted with the Smart Valve Interface (SVI[®]) positioner/controller.

depending on piston position and velocity in the water area, without touching the metallic valve parts.

To enable a control valve to fulfil its duty, i.e. continuous throttling of the rate of flow, the valve must be properly dimensioned. By dimensioning, not only the sizing of the valve is meant but also the adaptation to the duty prescribed, taking into account the specific operational conditions including assessment of the cavitational behaviour. The cavitational behaviour of a valve can be observed in a model test from which the behaviour in the actual installation can be deduced.

As a means of comparison, the cavitation coefficient σ , also known as the Thomas cavitation number, has been introduced. This value indicates the start of cavitation. The cavitation number δ is calculated from:

$$\delta = \frac{p_2 + p_A - p_D}{p_1 - p_2 - \frac{V^2}{2g}}$$

where p_A = atmospheric pressure

p_2 = pressure downstream of disturbance

p_D = vapour pressure of water

p_1 = pressure in undisturbed upstream side

V = velocity in undisturbed upstream side

g = acceleration due to gravity

Responsible valve makers determine the behaviour of their valves in all-embracing tests and therefore possess comparison values for all common operating conditions. From these data, the required design shapes can be deduced. In extreme cases for which test data are not yet available, the particular case is reconstructed in model tests and the required design determined. The main precondition is precise knowledge of the operating and installation conditions at the project stage.

Only with these data is it possible to determine the optimum design of the valves and piping run with the aim of avoiding damaging cavitation effects. Among the necessary details are knowledge of:

- (i) static and dynamic pressure upstream and downstream of the valve referred to the desired range of rates of flow
- (ii) the installation situation within the pipeline
- (iii) the piping and built-in components downstream of the valve
- (iv) the maximum permissible head loss, etc.

A close co-operation between designer, user and manufacturer of valves has in the past always proved to be very beneficial, and will doubtless lead to equally satisfactory solutions in the future.

Noise Control

Noise produced in pipelines may be pump-generated (changes in power and pressure, or varying amplitudes of pressure pulsations) or fluid-generated (flow instability, turbulence or simple fluid friction).

Fluid-generated noise in small-bore pipes with low to moderate flow rates is generally negligible, unless pressure pulsations are present (e.g. owing to valve cavitation). Thus pipe vibration, and consequent radiation of airborne noise, is usually caused by a higher level of noise generated by fittings; pipe resonance is caused by a mechanical vibration or resonant noise generated in supporting systems.

Relative noise levels are shown in Table 1.

Specifically, noise control (noise abatement) falls into two distinct categories:

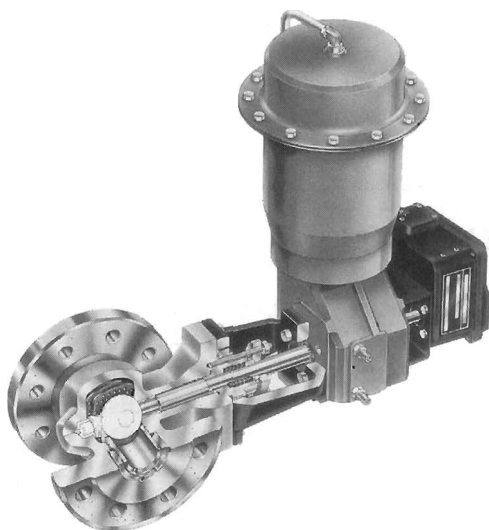
Source treatment: i.e. design of components to ensure operation at minimum noise levels.

Path treatment: to reduce source-generated noise to acceptable levels.

Noise caused by the operation of valves, regulators and control elements is transient and related to the degree of turbulence or cavitation produced although, in specific designs and certain circumstances, individual elements may be subject to vibration and generate a continuous noise. So much depends on the design and finish of the flow passages involved that no general analysis can be attempted. The noise level of such devices is dependent on the design and the localised flow velocities produced, and also on the response

Table 1. Relative noise levels

130 decibels	Jet aircraft on take-off
120 decibels	Threshold of feeling
110 decibels	Elevated train
100 decibels	Loud highway
90 decibels	Loud truck
80 decibels	Plant site
70 decibels	Vacuum cleaner
60 decibels	Conversation
50 decibels	Offices



Rotary eccentric plug control valve with noise control trim.

time, where applicable. The latter effect can be minimised by making sure that the response time is not shorter than that required by the system. This will result in minimum 'hammer'. 'Water hammer', in fact, depends on the switching velocity of the valve, e.g. on the spool-switching velocity in the case of spool valves. Valves operated by dry solenoids have uncontrolled response and so often produce 'hammer'. Wet solenoids are cushioned by the fluid so move more smoothly and open the valve passages more gradually (at the expense of some loss of solenoid power).

As a general recommendation, simple undamped ball-and-spring non-return and relief valves should not be used. On the design side, every effort should be made to ensure that the flow passages of valves are swept and free from sharp edges and corners as far as possible. Directional control valves must also be carefully designed to prevent flow instability occurring.

About 80% of the noise problems in process industry control valves are caused by flowing gas and 20% by flowing liquid. The noise caused by liquid is more often associated with cavitation, corrosion erosion and vibration.

Noise prediction has been made a lot easier by virtue of a number of manufacturers' software programs that have become available for general use.

Source treatment

Source treatment is difficult to describe in general terms because it is mainly concerned with the design of optimum flow paths through valves to reduce or eliminate noise that would otherwise be generated. In this respect, quite

different design parameters are involved in dealing with aerodynamic noise resulting from liquid flow.

Aerodynamic noise

Noise generated in gas or vapour is called aerodynamic noise. Most of it occurs during the deceleration stage in the throttling process. The area where the noise is generated can extend a long way from the orifice into downstream piping. Pressure waves inside the piping make the wall vibrate.

Noise is attenuated very slowly in piping filled with gas or vapour. The sound pressure level of a gas control valve generally has a broadband frequency distribution. Maximum sound pressure levels are between 1000 and 4000 Hz. There are a number of methods of diminishing aerodynamic noise. Two, in particular, are effective:

- (i) reduction of pressure and velocity gradients generated during the throttling process;
- (ii) using, for example, multi-stage throttling and splitting the flow into several jets.

The mean flow velocity and its profile downstream of the valve have a particularly marked effect on the valve noise level. Splitting the flow into smaller parallel jets reduces noise.

A typical frequency distribution of aerodynamic noise is shown in Figure 1.

Two examples of aerodynamic noise treatment are shown in Figures 2 and 3, applicable to globe- and angle-valve bodies. Both are cage-style valves, one using a cage with multiple slotted orifices of special shape, size and spacing, and the other a cage with multiple hole orifices. Claimed performance is an 18 dB reduction for the former compared with a conventional valve of similar type, and a 30 dB reduction for the multi-hole orifice cage. The latter is also particularly effective for applications involving high differential pressures (pressure drop across the valve). A common feature of both these valves is an expanded outlet design to minimise regeneration of valve noise.

Hydrodynamic noise

When turbulent liquid flow is stable, it does not usually cause any significant noise. Cavitation is the most common cause of noise in liquid flow.

Hydrodynamic noise can be reduced by affecting the intensity of cavitation. The best way to prevent cavitation is to intensify flow losses, which reduces the intensity of pressure recovery and increases the acoustically determined differential pressure ratio of incipient cavitation. Valves can be designed so as not to direct any cavitation jets at the valve trim; this helps to lower the effect of cavitation corrosion.

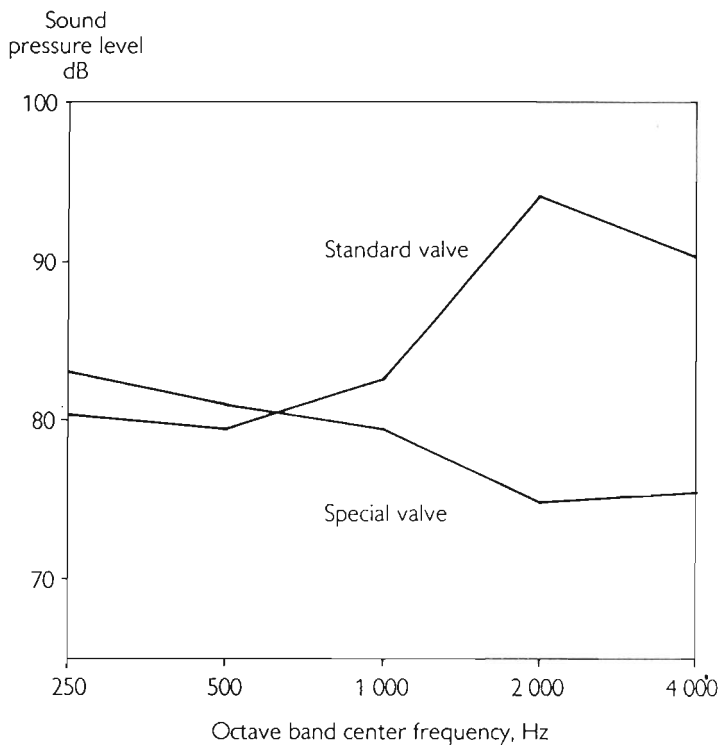


Figure 1. Typical frequency distribution of aerodynamic noise.

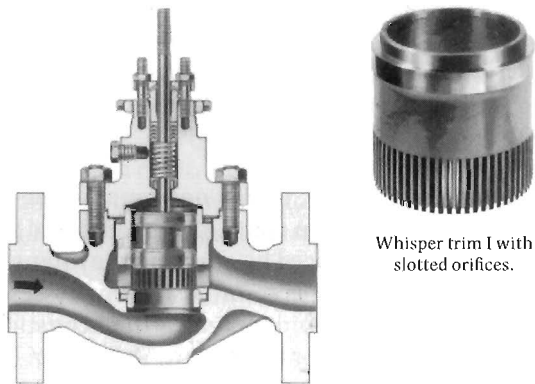
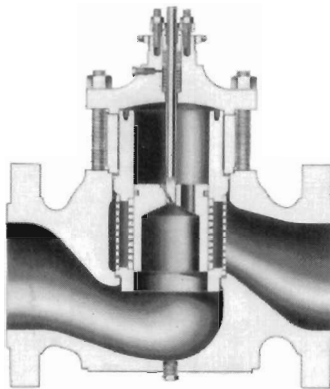


Figure 2. Trim cage and valve-body assembly.

Examples of cage-type valve trims for hydrodynamic noise treatment are shown in Figures 4 to 7. Here, the immediate aim is to eliminate or minimise cavitation. The cage design of Figure 4 uses one stage of diametrically-opposed flow holes through the cage wall to reduce both cavitation noise and damage. Each specially-shaped hole directs a jet of cavitating liquid which impacts with the jet admitted from the opposing hole at the centre of the cage. Thus, a continuous cushion is formed which prevents cavitating liquid from contacting the metal surfaces and ensures that vapour-bubble collapse takes place in the centre of the flow stream.

The cage design in Figure 5 consists of one or more concentric cylindrical sections referred to as stages. The number of stages required depends on the inlet pressure and the pressure drop. In operation, the liquid undergoes a portion of the total pressure drop in each stage of the cage. This prevents the liquid in any one stage of the cage from falling to or below its vapour pressure. Therefore, formation of vapour bubbles and their subsequent collapse is eliminated.

Figure 6 shows a further trim design employing a (patented) pressure staging for elimination of cavitation with differential pressures above 200 bar (3000 lb/in²). The expanding flow area design takes advantage of the ability



Whisper trim III cage in Fisher
EWD control valve-body
assembly.



Whisper trim III cage.

Figure 3.

of the liquid to undergo a greater pressure drop in the initial stages without cavitating. This results in a much lower inlet pressure to the final stage. This design also separates the shut-off and throttling locations to prevent clearance-flow erosion.

A further design is shown in Figure 7 where the trim consists of a carefully designed bundle of tubes which minimises cavitation noise and damage by controlling the formation of cavitation-bubbles. The tubes serve three functions: they prevent the flow stream from reaching its potential minimum area, they maintain maximum pressure head to reduce cavitation bubble formation, and they limit the size and number of cavitation bubbles that do form.

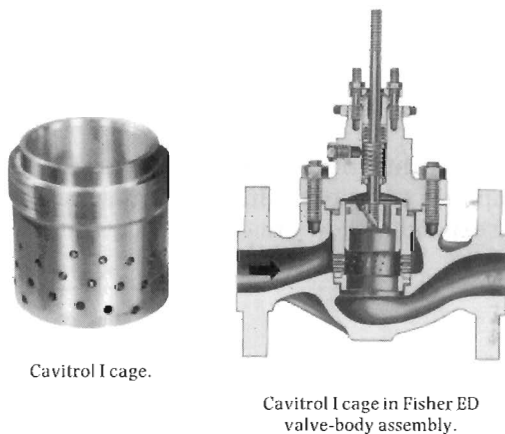


Figure 4.

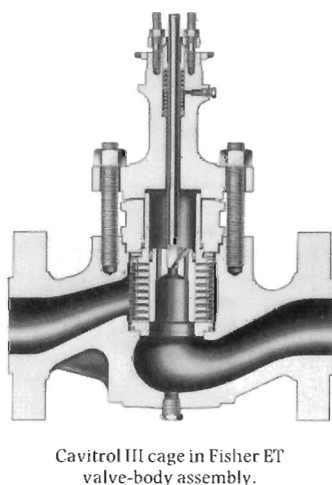


Figure 5.

Ball valves

Rotary-control ball valves were previously noisy due to the high recovery character of the ball valve and cavitation at high differential pressures.

The first low-noise anti-cavitation ball valve was introduced in 1979. It was based on a multi-stage, multi-flow principle, with a trim of variable resistance depending on the valve opening.

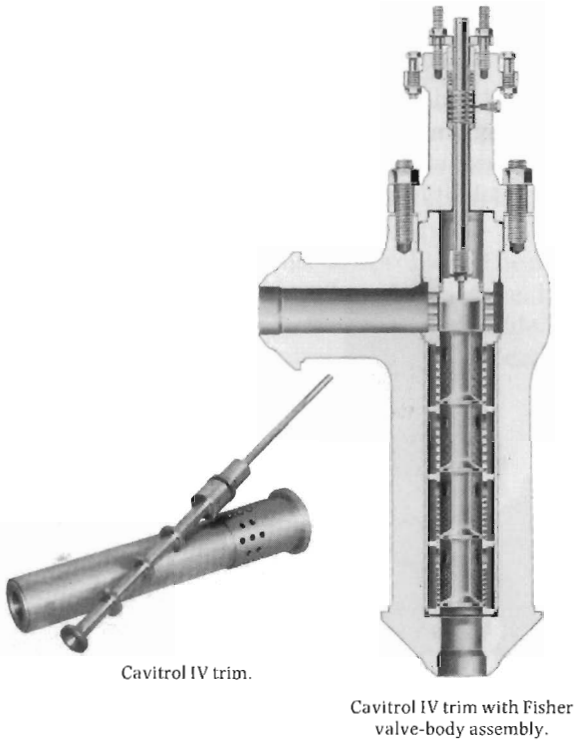


Figure 6.

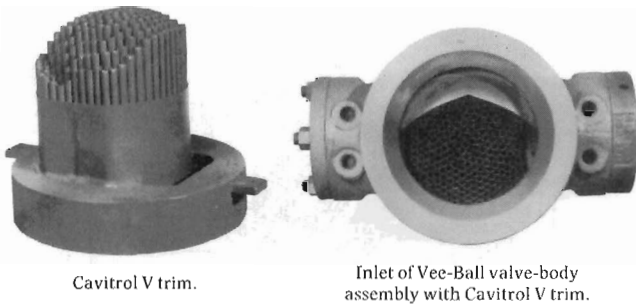


Figure 7.

The quiet metal-seated control-ball valve shown in Figure 8 combines the concept of low noise with minimum cavitation.

Parallel perforated plates in the ball flow opening smooth the pressure drop as the flow passes through. The gradual pressure reduction over the valve reduces velocities, noise generation and cavitation.

When the ball is opened, the fluid passes through the upstream seating orifice encountering resistance inside the ball flow opening. The flow is forced through the holes in the perforated attenuator plates (Figure 9).

The plates create a frictional path, where each plate and the seating orifices reduce the pressure step by step. This prevents excessive velocity generation, lowers the noise level and minimises cavitation.

When the opening angle is increased, resistance decreases as the flow successively by-passes the plates. This gives optimal valve-flow characteristics and thus high rangeability and capacity.

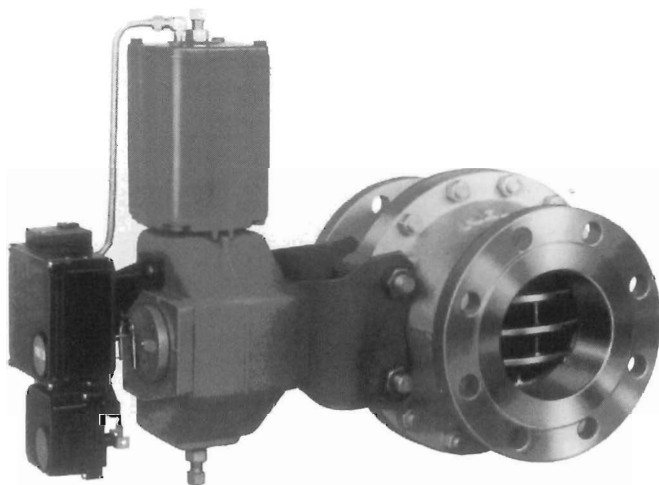


Figure 8. Quiet metal-seated control ball valve.

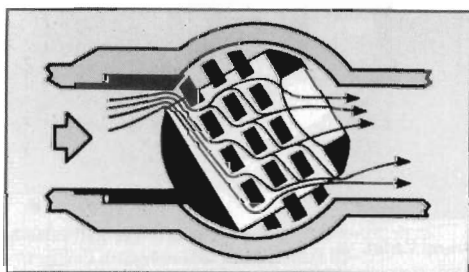


Figure 9. Principle of the ball with parallel perforated plates.

This type of valve has proved successful in many applications industries including:

- hydrocarbon processing
- power generation, chemical, and pulp and paper industries
- flow and pressure control, especially in critical flow conditions
- blow-down
- pressure equalisation
- high-temperature service and tight shut-off requirements

Butterfly valves

With conventional butterfly valves, increasing differential pressure causes a high dynamic torque, thus jeopardising controllability at high opening angles and causing noise with gases and cavitation with liquids.

The valve shown in Figure 10 overcomes this problem. The non-symmetrical pressure-distribution pattern on both sides of the vane has been made symmetrical with a downstream partial flow obstacle inside the valve body. This design helps to eliminate the dynamic torque and, because of the more turbulent flow pattern, lowers the recovery behaviour.

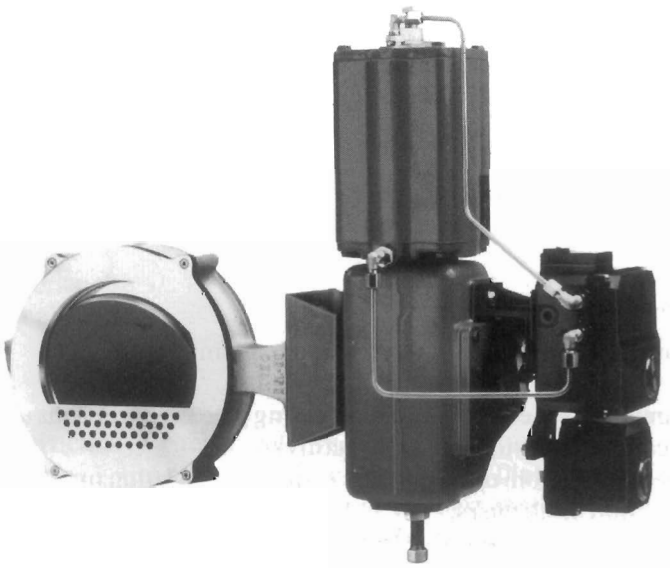


Figure 10. 'S' disc butterfly control valve with flow-balancing and noise-control trim.

The disc has been designed to remove fluid forces from the disc to the body and the 'flow-balancing trim' has been incorporated to optimise the inherent flow characteristics of the valve. As a result, noise and vibration are reduced.

The valve is used in many process applications within the temperature range -200 to $+700^{\circ}\text{C}$ (-333 to $+1300^{\circ}\text{F}$).

Pressure-relief valves

The following formulae are used for calculating noise levels of gases, vapours and steam as a result of the discharge of a pressure-relief valve.

The expressed formulae are derived from API Recommended Practice 521.

$$L_{100} = L + 10\text{LOG}_{10}(0.29354 W k T/M)$$

Where L_{100} = sound level at 100 ft from the point of discharge (decibels)

L = noise intensity measured as the sound pressure level at 100 ft from the discharge

W = maximum relieving capacity (lb/hr)

k = ratio of specific heats of the fluid (for steam, $k = 1.3$ if unknown)

T = absolute temperature of the fluid at the valve inlet ($^{\circ}\text{Rankine}$ ($^{\circ}\text{F} + 460$))

M = molecular weight of the gas or vapour obtained from standard tables

When the noise level is required at a distance of other than 100 ft, the following equation should be used:

$$L_p = L_{100} - 20\text{LOG}_{10} (r/100)$$

where L_p = sound level at a distance, r , from the point of discharge (decibels)

r = distance from the point of discharge (ft)

Path treatment

Standard methods used for noise reduction in piping are:

- (i) Damping by means of suitable isolating pipe supports. This also provides decoupling for supporting structures.
- (ii) Decoupling from other sources of noise or vibration in the system.
- (iii) Insertion of silencers.
- (iv) Soundproof 'lagging'.
- (v) Use of bearing-walled pipe.

For the majority of systems, only (i), and to a lesser extent (ii), should be necessary. 'Lagging' is normally only required when there are pulsation

vibrations present which cannot be damped or isolated by simple means. This is most likely to occur in pumped systems employing thin-walled, large-diameter piping, particularly on the suction side.

Sufficient damping for pipes is usually provided by suitable supports, or pipe clips, spaced at regular intervals, the supports having resilient linings so that vibration in the pipe is not transmitted directly to the surface to which the supports are fixed.

Optimum pipe spacing can be analysed in terms of standing wave phenomena, although this is seldom necessary. The case of axial standing wave is usually academic, for practical lengths are usually substantially lower than the critical length, which is defined by:

$$L_a = \frac{8200}{f} \text{ for steel pipes}$$

where L_a = resonant length of pipe (ft)
 f = frequency of any strong vibration

Theoretically, at least, the distance between pipe supports should always be less than this resonant or critical length.

It may, however, be necessary to analyse the various possible sources of noise in a fluid pipework system in more detail in order to arrive at satisfactory noise treatment. In this case, the possible sources of noise generation, in decreasing order of significance, are:

- (a) pump noise (where applicable)
- (b) appliance noise
- (c) control element noise
- (d) water hammer
- (e) chatter
- (f) cavitation
- (g) resonance
- (h) pipework noise
- (i) thermal effects

Bellows

Bellows—particularly rubber bellows—can be very effective in preventing pump noise from being transmitted along pipelines. Plain elastomeric bellows provide a complete isolation joint and give the best possible sound absorption as all pipe-borne noise must pass through the bellows material. For best results, the bellows should be placed as close to the pump as possible, and the pipe to which it is connected securely anchored as near as possible to the other end of the bellows. The pump also needs to be solidly mounted to withstand both pressure forces and flexibility forces arising out of the bellows' stiffness.

Where this is not possible (e.g. the pump is flexibly mounted), or other factors (such as high pressure) mitigate against the use of plain bellows, tied or axially-restrained bellows must be used. Such bellows are pressure-balanced units. The flanges and tie-bars, however, now form a transmission path for vibration unless isolation treatment is incorporated. The simplest form of treatment is by the use of resilient bushes and/or rubber washers to prevent metal-to-metal contact between the tie-bars and the backs of the restraining flanges. Even single rubber washers can be effective, if correctly selected in terms of hardness.

Some noise-reduction data obtained with representative designs are shown in Figure 11.

Metal bellows

Metal bellows can give inconsistent results in terms of noise and vibration isolation. Generally their performance is much below that of rubber bellows. Again some test data are shown in Figure 12.

Isolating flanges

An alternative type of isolator is shown in Figure 13. Basically, this consists of a solid rubber 'washer' of appreciable thickness, into which are bonded steel flanges. These flanges are tapped to accommodate bolts for assembling the

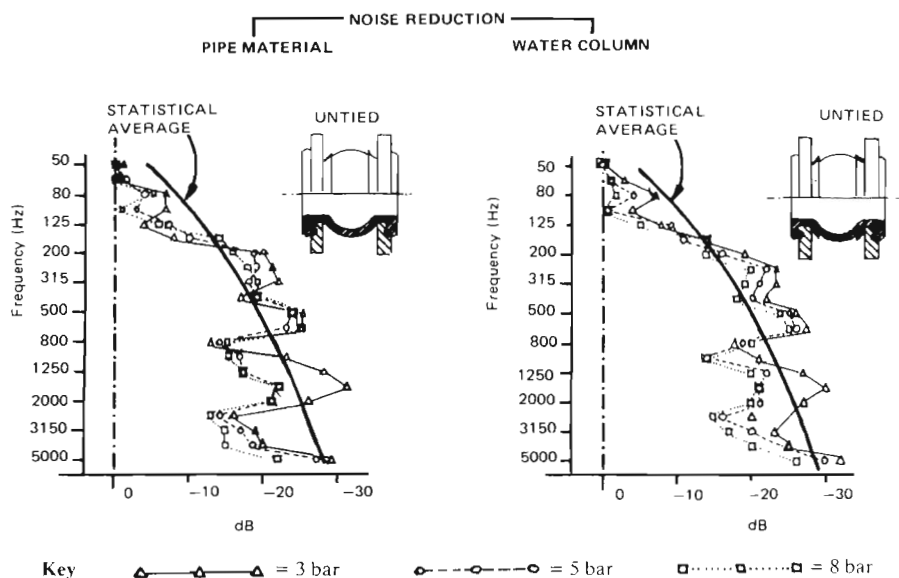


Figure 11. Typical noise reduction with simple rubber bellows.

isolator between conventional flanges without metal-to-metal contact through the joint.

Theoretically, such a form of isolator should prove better at higher frequencies than lower frequencies, although actual performance would depend on the hardness of the elastomer used. It is not generally as effective as rubber bellows, and definitely inferior for isolating lower frequencies. It does, however,

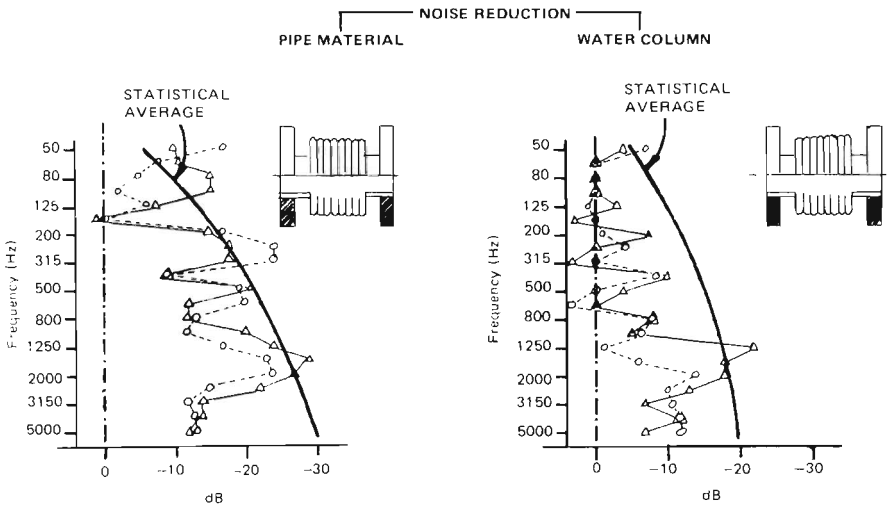


Figure 12. Typical noise reduction with corrugated-steel bellows.

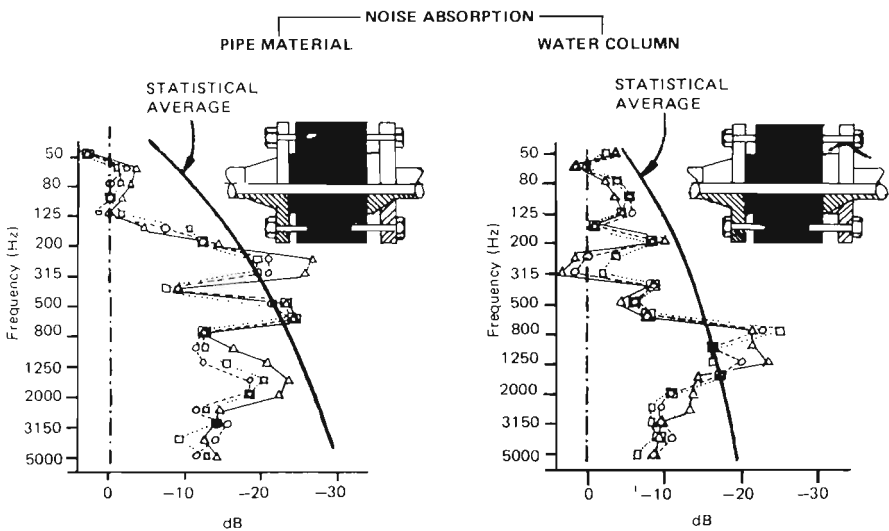


Figure 13. Solid metal/solid rubber isolating unit and typical noise-reduction data.

have the advantage of providing a 'solid' coupling and so can be used with flexibly-mounted pumps.

Acoustic filters

Acoustic filters can be fitted to systems where pressure ripple is high. These are essentially tuned silencers which are critical in design and are usually effective over only very narrow frequency bands, although the attenuation achieved can be quite high. Untuned silencers simply comprise an expansion chamber with broader coverage but reduced attenuation. An accumulator is, in effect, an untuned hydraulic acoustic silencer and is most effective at lower frequencies. Dissipative-type silencers provide for dissipation of energy through viscous flow losses and, as a consequence, consume some fluid energy. They may be combined with an untuned silencer, although the attenuation will still be appreciably lower than that of the tuned type.

In general, wave-cancelling filters are to be preferred because the frequencies involved are low. If the pressure transients are narrow band, a Quinke tube and expansion chamber can be effective (Figure 14). A major disadvantage of this and other types of simple wave-cancelling filters, however, is the relatively high pressure drop produced. The more usual form of hydraulic silencer is the pressure-release type shown in Figure 15. This gives minimum

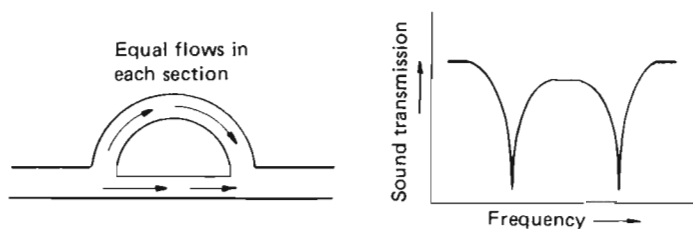


Figure 14.

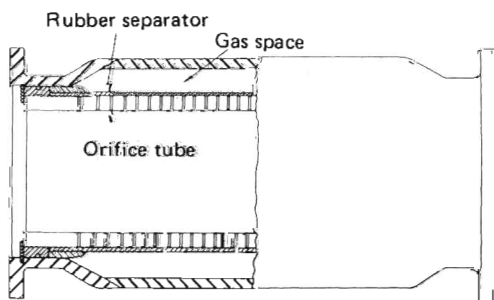


Figure 15. Pressure-release filter.

pressure drop and broad-band filtering, but is pressure-sensitive and needs regular routine maintenance.

Shock preventers

Shock preventers are pulsation dampers (or accumulators) characterised by having very large flow-inlet apertures which are partially closed off by liquid trying to flow back out of them. They are not shock absorbers, as they prevent shock or surge occurring. For the same reason, they do attenuate shock.

Shock removers

These are sensitive hydropneumatic devices which prevent a standing wave from passing farther down a system or from bouncing back through them. They are normally of tubular or sleeve form with a flexible membrane. Because of their length, it is possible to open a membrane so that it is exposed to the increased pressure of a wave and to close it behind the wave, thus shutting it in.

See also the chapters on Cavitation and Flow of Liquids Through Pipes.

Balancing of Hydronic Systems

Excess consumption of heating energy is the result of temperature variations in a building and the generally incorrect approach taken to solve the problem. If, for example, one room is too warm and another too cold, the necessary adjustments are often made in the room that is too cold—the result being that the average temperature of the building, which to start with was probably quite sufficient had it been balanced out, now becomes too high (see Figures 1 and 2).

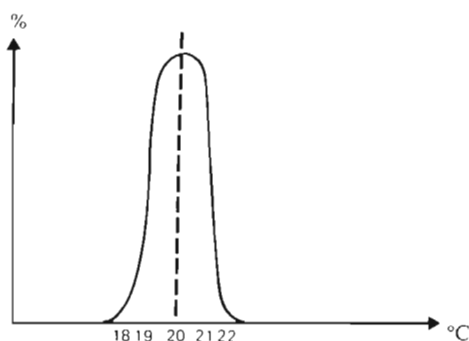


Figure 1. Example of correctly balanced building. Average temperature 20°C.

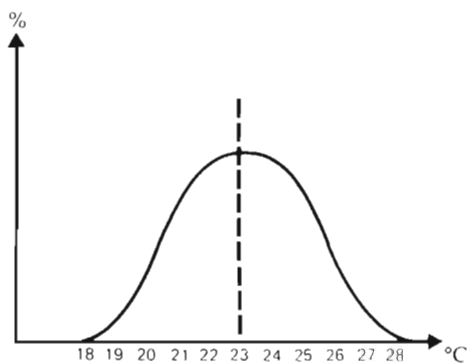


Figure 2. Example of wrongly balanced building. Average temperature 23°C.

Obviously, this higher average temperature implies a higher consumption of energy; how much higher depends on the specific heat requirement of the geographical area in degree/days. Generally speaking, however, one can say that each degree above $+20^{\circ}\text{C}$ ($+68^{\circ}\text{F}$) indoors means heating costs which are about 5 or 6% higher than normal (see Figure 3).

In a cooling plant the conditions and the means are of course reversed, but the waste of energy is often increased.

In many cases, excess temperature is even ventilated away, or the windows are opened, which can mean an average temperature which is more than $4\text{--}5^{\circ}\text{C}$ ($7\text{--}9^{\circ}\text{F}$) too high.

Energy-conservation measures have always been of considerable interest and the balancing of a hydronic system is particularly interesting because only simple measures are needed and they give quick and very evident results. Savings of 20 to 30% are not unusual.

Pump-energy waste

It is often forgotten that pump energy also costs money. In many cases, the pumps are oversized.

In heating systems, this is not always so very significant because the temperature differences are often high and, therefore, a relatively small amount of water is being circulated. However, particularly in refrigeration systems with their lower temperature differences, pump-energy waste can add a lot to the operating costs.

Another essential difference between heating and cooling systems is that energy losses in the systems are converted to heat; this works to the benefit of a heating system but necessitates an increase in capacity for a cooling system.

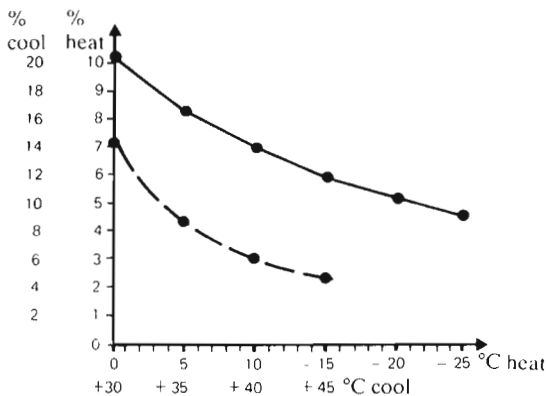


Figure 3. Solid line: Energy cost saving in % for each degree higher indoor temperature at various maximum outdoor temperatures in a cooling system. Broken line: Energy cost saving in a heating system in % for each degree lower indoor temperature at various minimum outdoor temperatures.

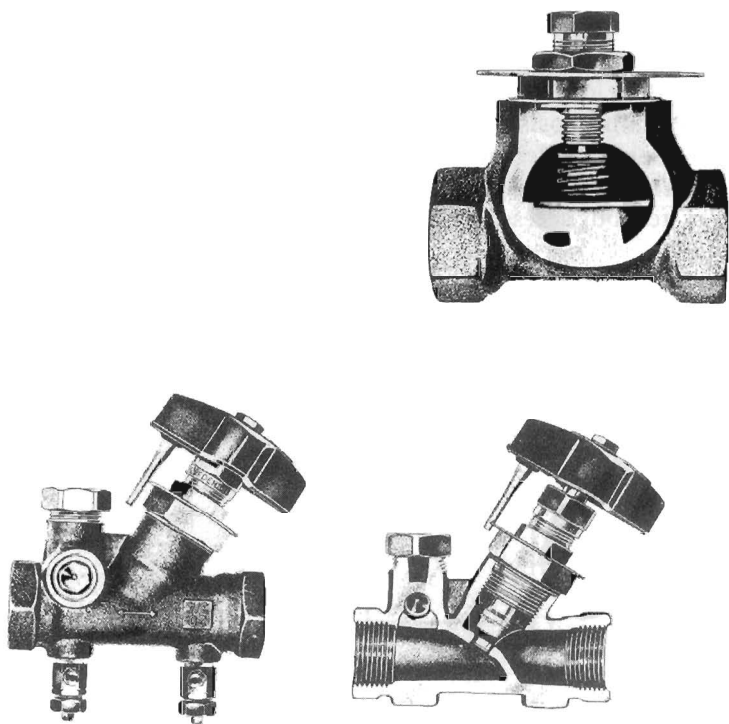
Over-dimensioning is generally a consequence of the following:

- (i) When the pump was chosen, the designer was uncertain of the pressure drop in the system (boiler, heating batteries, valves, etc.) because the tender covering the components had not been finally accepted.
- (ii) Insufficient data concerning the pressure drop in the piping system.
- (iii) General safety factors.

However, to facilitate balancing, it is wise to increase the sizes of the pump slightly, but not to exaggerate.

Modifications to an over-dimensioned system (for example, by changing the pump or the impeller), do not always lead to lower overall costs, although such measures do often prove worthwhile.

For some time now, it has been taken for granted that a hydronic system in a new building must be balanced. The question being discussed is how to balance it and who is to balance it—the application engineer, the heating contractor, or a firm specialising in the balancing of hydronic systems.



Examples of balancing valves.

Construction

To skip the piping calculations entirely and simply specify that the system shall be balanced without detailing the means or the way to do the work implies a significant amount of extra work for the person who is to perform the balancing. Furthermore, balancing is not a universal solution which will make a poorly designed system function adequately.

When designing the system, care must be taken to arrange clearly demarcated sections.

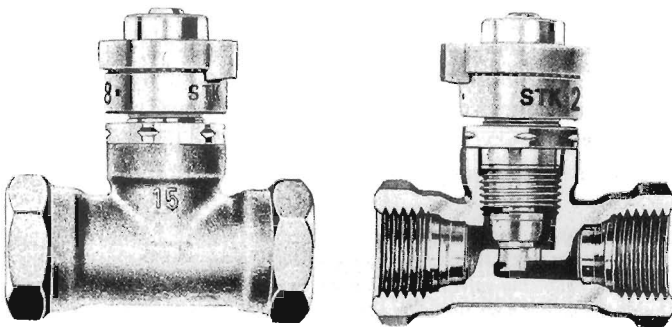
Calculations

For newly-built facilities in Sweden, for example the SBN 80 39:32 specifies that the method of balancing and the presetting values and water-flow values for the balancing valves must be detailed on the building permission documents. This necessitates complete piping calculations and the determination of pressure losses in the heating system.

The K_v values can, in the case of radiator valves, generally be adjusted directly by means of the presetting unit which is marked off in K_v , although this only applies in a limited number of manufacturers' valves.

Balancing valves must be incorporated in all branches to avoid having to balance the radiator valves in one branch against the balancing valves in other branches (Figure 4).

Reverse-return mains, according to the Tischelmann system, can simplify many balancing problems. The exclusion from such a system of balancing valves will, however, lead to imbalance, because different radiators and heaters or cooling units do not have the same output or pressure drop. However, the pressure differences in the system between branches will be lower, which means that balancing will be easier. The Tischelmann reverse-return system also has benefits to offer when balancing the sub-circuits (Figure 5).



Balancing and shut-off valves.

Balancing

Before commencing to balance a system, all valves must be opened fully. This applies particularly to thermostatic radiator valves and two-way control valves. This type of valve operates with varying flows and, unless it is ensured that the valve is fully opened, it may just have closed automatically as balancing was commenced.

A thorough knowledge of the system is also important before the commencement of balancing. The information required includes the following:

- (a) drawings with hydronic sketches
- (b) data concerning flows and pressure drops across heat generator, batteries, radiator heaters, balancing and control valves
- (c) pumps data and pump diagram.

The desk method

A well-defined system of not too great complexity generally requires only one single adjustment of all valves used, including radiator valves and balancing valves, in accordance with the values specified by the drawings. Control measurements should, however, be made on one or more extremity branches

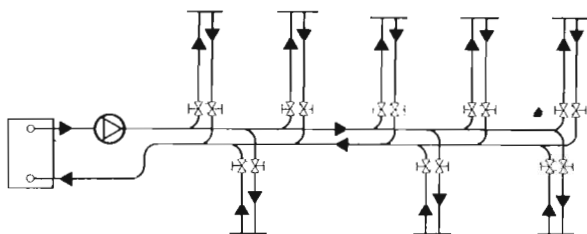


Figure 4. Balancing valves incorporated in all branches to avoid having to balance the radiator valves in one branch against the balancing valves in other branches.

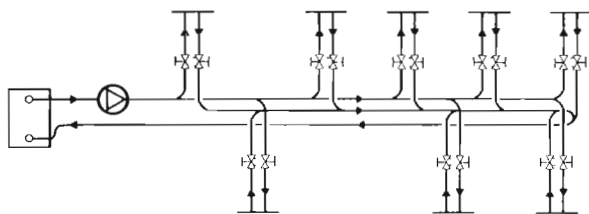


Figure 5. The Tichelmann reverse-return system can simplify many balancing problems. The exclusion from such a system of balancing valves will, however, lead to imbalance, since different radiators and heaters or cooling units do not have the same output or pressure drop, but the pressure difference in the system between the branches will be lower. This means that balancing will be easier. The Tichelmann reverse-return system also has benefits to offer when balancing the sub-circuits.

(farthest away from the pump) as well as on a few central branches. The flow deviations found in these control measurements should not exceed 10% of the volume or 20% of the pressure. If, after an adjustment as described earlier, there are still temperature deviations of more than 1°C (2°F) in individual rooms, and if these cannot be related to temporary fluctuations in the heating or cooling load, the explanation will be found in one of the following: incorrect calculation of the heating/cooling facility; incorrect design/installation of heating/cooling facility; bad building (insufficient sealing, draughts).

Temperature measurement method

This method, which is only applicable to heating systems, is based on the fact that each radiator/heater is dimensioned according to the same temperature drop with an equal outdoor temperature. As a consequence, the system can be balanced by measuring the temperature drop at the pump and then adjusting the balancing valves so that the temperature drop is the same at the pump as it is over each branch.

To achieve acceptable accuracy with this method, the outdoor temperature must be almost constant throughout the entire balancing process and, in addition, below 1°C (34°F). It is often small temperature drops that are being measured and therefore the temperature differences become even smaller. For this reason, the system is at times less than exact (see also Figure 6).

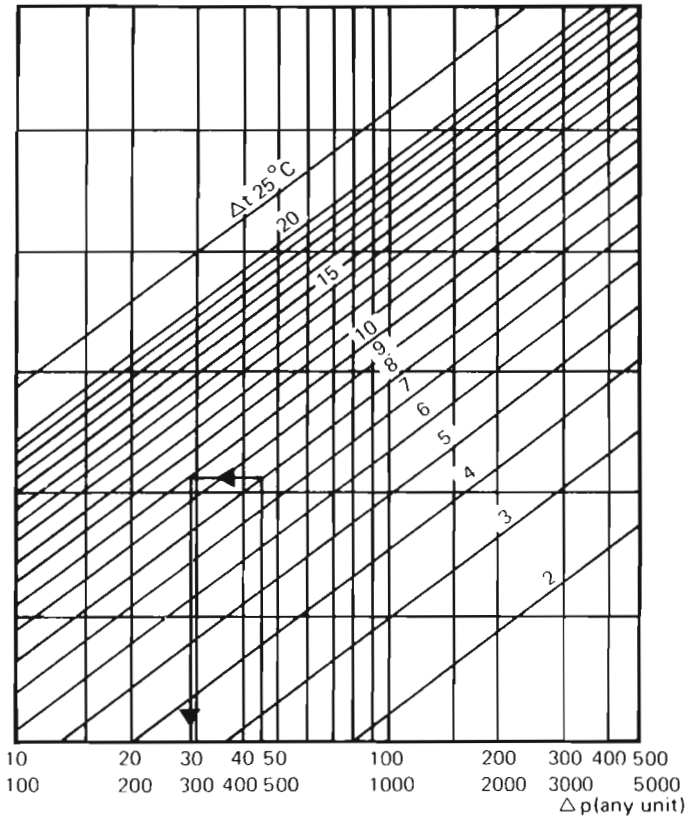
The temperature method can also save time if it is used as a preliminary stage prior to the proportionate balancing method described below.

Proportionate balancing method

This method is one of the most frequently used and it is suitable for old facilities as well as for new ones.

The procedure is to measure the pressure drop and to move proportionally from branch line to branch line. This is done as follows:

- (1) Set all radiators and balancing valves according to the drawings specifications—or open them if no specifications are available (Figure 7).
- (2) Start by measuring branch lines 1.1, 1.2, 1.3 and 1.4 of sub-system 1 and determine the proportionate flowrate, that is to say the relationship between measured flow rate and design flow rate. If flow in 1.1 is 1500 l/h and the design flowrate is 1000 l/h, the proportionate flow rate will be 1.5.
- (3) Presume that 1.1 has the lowest proportionate flowrate, 1.2 the next lowest and so on. In this case, leave the valve of branch 1.1 open and balance 1.2 to give the same proportionate flowrate as 1.1 (within the tolerance applying). These two branches are now balanced and you can continue with 1.3, balancing it against 1.2 until they both have the same



Nomographic chart for the balancing of branch lines on the basis of combined pressure and temperature difference measurements.
In the example, the measured pressure difference is 45 mm Hg and the temperature difference is 8°C .
If a 10°C difference is desired the balancing valve must be throttled to give a pressure difference of 29 mm Hg.

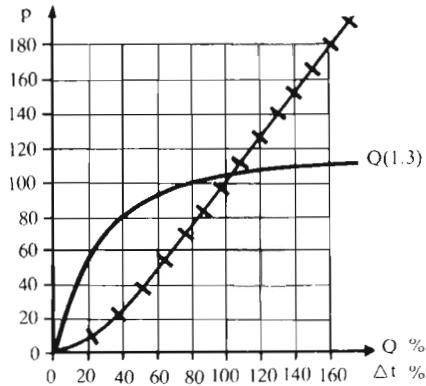


Figure 6. The heat emission variations in % as a function of temperature changes Δt in % and flow changes in % (an 80–60 radiator system). The diagram shows that a 1% deviation in temperature will give a significant deviation in heat emission. A deviation in flow will influence heat emission to a much lesser extent.

proportionate flowrate. 1.1 need not be checked. It will change in direct proportion to 1.2 and remain in balance with 1.2. Then continue with 1.4 in the same manner.

Because 1.4 is the last branch line of sub-system 1, this means sub-system 1 is now ready and, if any flow changes occur in the total system, the branch lines of sub-system 1 will be altered by the same proportional amount to a new common proportionate flowrate.

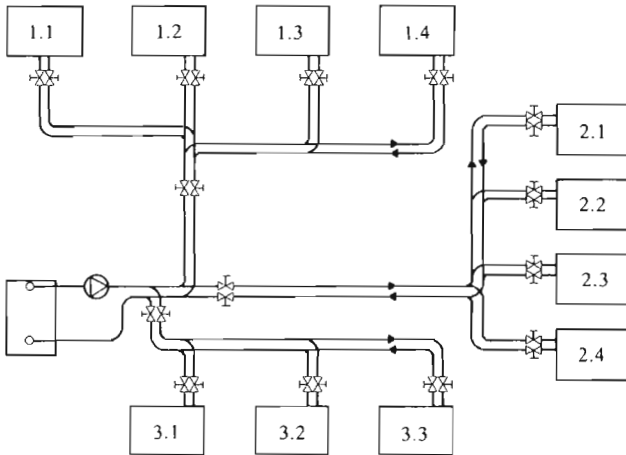


Figure 7.

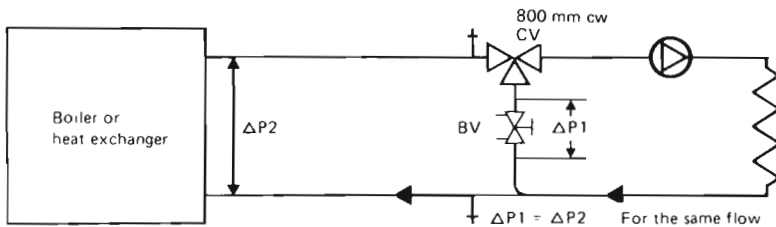


Figure 8. Mixing valve.

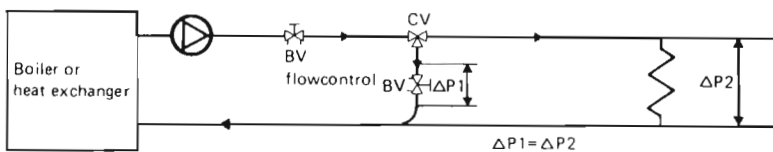


Figure 9. Diverting valve.

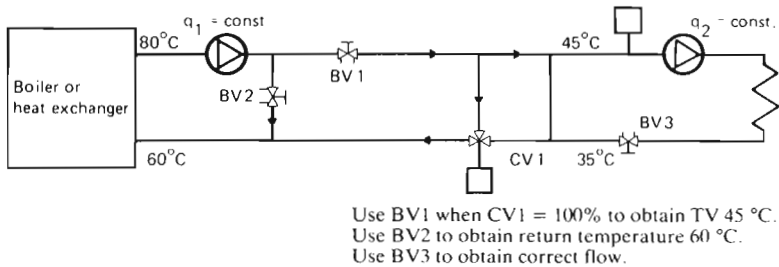


Figure 10. Double mixing valves.

- (4) Proceed in a similar manner with sub-systems 2 and 3.
- (5) Leave the balancing valve open in the sub-system which has the lowest proportionate flowrate and balance the other sub-systems as described earlier.
- (6) The final stage is to determine whether or not the pump is supplying too large a quantity of water. If it is giving too much water, it can be throttled by a means of the balancing valves, by altering the pump speed, or by changing the impeller. In the case of large oversizes, it is often best to reduce the speed of the pump or change the impeller.

Some other examples are given in Figures 8–10.

SECTION 8

Duties and Services

Water Services

Hygienic Services

Steam Services

Fire-Safe Valves

Fire Hydrant Valves

Marine Services

Vacuum Services

Cryogenic Valves

Nuclear Services

High Pressure Services

Water Services

On a global basis, the valves produced to handle water are generally made from cast or ductile iron or cast steel. They are, in the main, larger in size than valves for other industries. The water industry can be divided into two main areas:

- handling clean water
- handling dirty water or sewage.

Clean water is normally handled with butterfly and gate valves. Butterfly valves typically have steel bodies and gate valves have ductile- or cast-iron bodies. Sewage tends to be handled with gate valves as, although butterfly valves have good sealing characteristics, when the valve is opened, the disk is still flat within the channel and thus presents an obstruction to solids. Specialist applications use knife-gate valves or wafer-butterfly valves. Plastic valves are not used in hot-water supply generally.

Other common valve types for water and waste-water include sewage combination air valves, cushioned swing check valves, hydraulically controlled air and vacuum valves, cone valves and air release valves.

Some water companies appear to have moved to gate valves with a resilient seat with mainly rubber materials rather than metal-seat valves. These valves are also replacing rising spindle-gate valves on grounds of cost. Rubber or plastic O-rings, including PTFE, are standard packing materials.

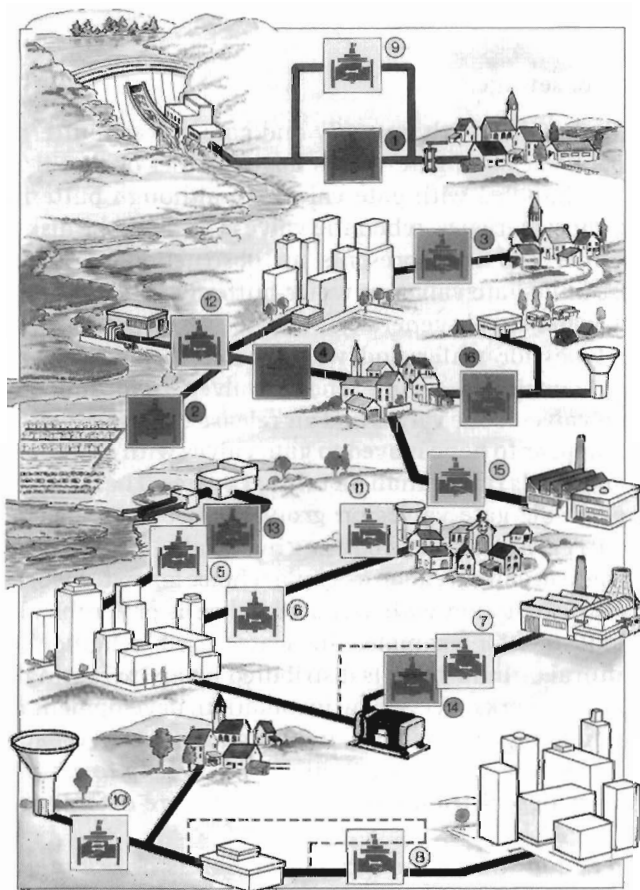
Metal-seated AWWA-type ball valves remain a popular choice for water-works and industrial specialities. Automation with actuated valves is preferable. In the water-distribution industry, for example, the water used, whether its purpose is urban, agricultural or industrial, is distributed by an increasingly complex pattern of pipeline networks. Every new installation, development or addition to the network (building development, industrial zone, etc.) creates an imbalance. Control valves in a water-distribution system help to restore the balance by directing water distribution according to pre-determined priorities.

It is important to understand that, although automation is now a major factor in the water sector, it is still limited as many valves within this industry are isolation valves that require to be operated manually.

The main requirement is reliability in operation with little or no disruption to the service.

Some common impurities in raw water are shown in Table 1.

The use of the plug valve for water-supply systems dates back to Roman times and is still widely used in its modern forms, together with various other types of rotary movement and screw-down shut-off valves. The main exception has been the ball valve. Only during the last 10 years or so has this type been developed to meet the technical and economic requirements of water-supply systems. In the meantime, various manufacturers have attempted to provide an alternative for drinking-water systems by offering 'intermediate' ('mixed') construction types, such as the segmented gate and ball valves. In practice, however, these types of constructions generally turned out to be hydraulically unstable and required very high actuating moments.



- ① Reduces pressure to a distribution system when gravity-fed from a source with a relatively high elevation.
- ② Reduces pressure in an irrigation system.
- ③ Maintains a preset upstream pressure and a preset downstream pressure reduction.
- ④ Reduces pressure to a low-pressure zone when the pump discharge is too high.
- ⑤ Prevents over-pumping of both deep-well and booster pumps if the system demand exceeds the pumping capacity.
- ⑥ Guarantees maintenance of sufficient upstream pressure when supplying water to a low-pressure zone.
- ⑦ Prevents a pump from lowering its suction pressure below a desired safe operating minimum.
- ⑧ Maintains a constant differential pressure across a pump to maintain a constant flow rate.
- ⑨ Protects the system against accidental overpressures (caused by a failure of the control valve C100, stop valve closing too quickly).
- ⑩ Double-action altitude valve allowing filling of the tank and emptying back to the supply.
- ⑪ Controls the level of the tank by means of float regulation and allows distribution to the village.
- ⑫ Protects the pump station against surges due to start-up, shut-down and power failure.
- ⑬ Eliminates pressure fluctuations when pump starts and shuts down.
- ⑭ Controls flow rate to the factory.
- ⑮ Allows flow between two distribution systems (example: feeding a water-storage tank for peak distribution time).

Typical control application examples.

Compared with the general class of shut-off valves, the ball valve offers the following potential advantages:

- (i) In the open position, it provides free passage of water in the supply system with a diameter equal to the supply connections.
- (ii) It gives an unimpeded flow profile without any distortion.
- (iii) It offers the smallest resistance to flow, that is, a very small loss in pressure over the comparable supply-line distances.
- (iv) It is completely adaptable.
- (v) The change in the cut-off from open position to closed position requires a minimal change in place.
- (vi) The precise shape of the cut-off guarantees a seal of great integrity.
- (vii) It offers a favourable ratio of weight to stability resulting from the design of the ball or hollow sphere which withstands the pressure.
- (viii) It has low installation costs.

Ball valves for water supplies

It was not until there were ball-valve designs which took into account the specific requirements of a drinking-water supply (such as resistance to the formation of deposits and acceptable hydraulic performance in intermediate positions), and the cost requirements (namely, amortisation of the high use of energy even at low flow rates and short periods of useful operation), that the use of ball valves for all aspects of drinking-water supply systems became a

Table 1. Some common impurities in raw water

Name	Symbol	Common name	Effect
Calcium carbonate	CaCO_3	Chalk, limestone	Soft scale
Calcium bicarbonate	$\text{Ca}(\text{HCO}_3)_2$		Soft scale + CO_2
Calcium sulphate	CaSO_4	Gypsum, plaster of Paris	Hard scale
Calcium chloride	CaCl_2		Corrosion
Magnesium carbonate	MgCO_3		Soft scale
Magnesium sulphate	MgSO_4	Magnesite	Corrosion
Magnesium bicarbonate	$\text{Mg}(\text{HCO}_3)_2$	Epsom salts	Scale, corrosion
Sodium chloride	NaCl	Common salt	Electrolysis
Sodium carbonate	Na_2CO_3	Washing soda or soda ash	Alkalinity
Sodium bicarbonate	NaHCO_3	Baking soda	Priming, foaming
Sodium hydroxide	NaOH	Caustic soda	Alkalinity, embrittlement
Sodium sulphate	Na_2SO_4	Glauber salts	Alkalinity
Silicon dioxide	SiO_2	Silica	Hard scale

subject of interest. In contrast to the design of a conventional ball valve with seat rings on both sides to form a seal, the principles of the rotary piston valve were borrowed as the basis for the design of a valve having flow around a ball section (Figure 1).

This type of circulation stabilises the flow through the bore of the ball and prevents the damaging effects of cavitation and disintegration which occur in the intermediate positions. The ring gap is used for the through-flow of the medium as soon as a point of constriction is attained at which the pressure loss in the central bore is greater than the pressure loss in the ring gap.

The basis for the stabilisation of the flow is, however, that the ring gap for circulation exhibits small transient changes in the surface, so that sudden changes in pressure or velocity can be avoided. The desired resistance of the ball valve to the formation of deposits, of the sort that might be formed from minerals and foreign particles such as sand, can be achieved in a design which makes use of a surface not requiring special finishing (Figure 2).

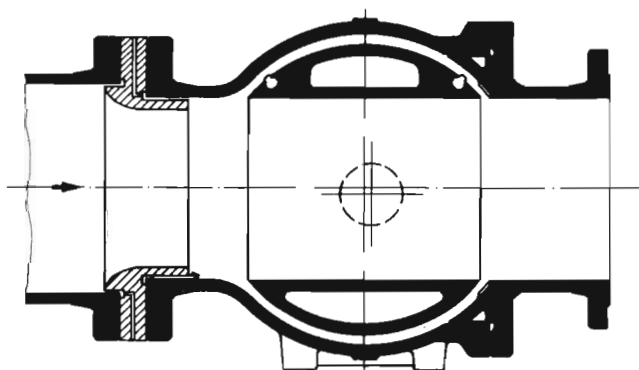


Figure 1. Ball valve: open position.

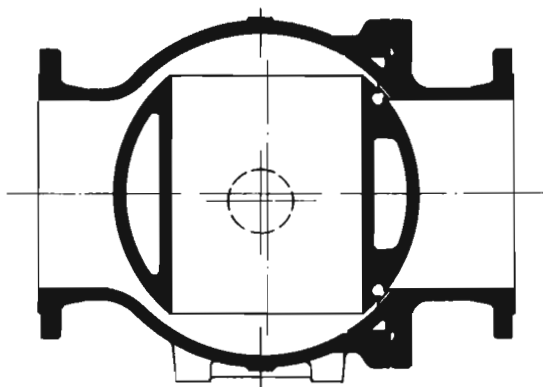


Figure 2. Ball valve: closed position.

This type of design requires the fixed placement of the seal elements, so that a rubber or an elastic preformed seal of a conventional variety would be first and foremost. To this end, there are already well-tested seals from the area of shut-off valves which have been around for decades and can be easily adapted. The counter seat in the housing can use a metallic seal made of corrosion-resistant steel with inlet and outlet edges having especially large radial sections which then provide a useful fixing in place of the rubber/elastic preformed seal.

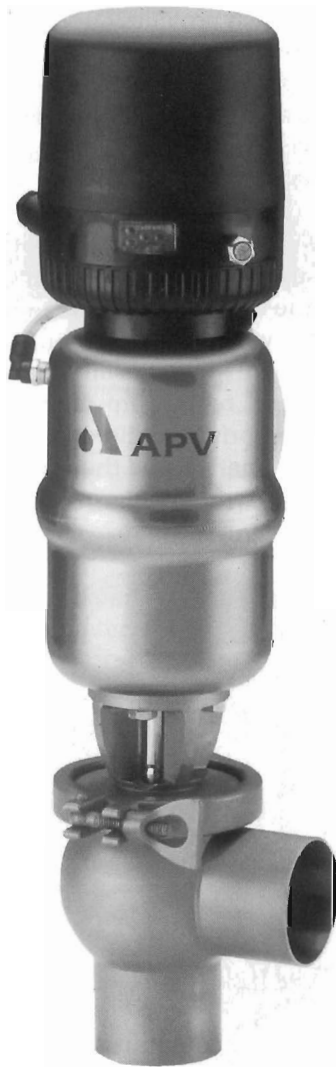


Figure 3. Hygienic service valve.

In addition, the bearing for the turning point of the ball is placed on an eccentric. This reduces the frictional load of these seal elements to the smallest levels, which is, of course, a necessary requirement. Mineralogical deposits and foreign particles can in this fashion block the ball in its entire periphery, so that the operating forces of the drive unit are not sufficient to move the ball.

For this reason, these types of ball valves are given a relatively large gap between the outer surfaces of the ball and the inner surfaces of the housing, to try to avoid gap corrosion—the formation of hard layers of deposits and corrosion. Additionally, the ball is provided with a scraper rim extending beyond the turning radius, which provides only a line contact and otherwise gives the ball surface free room in which to turn.

As a consequence, even with a large amount of deposits, the operating forces are sufficient to actuate the valve. In intermediate positions of the ball valve, this overall ring gap results in a washing action, and the medium moving through the valve produces a kind of self-cleaning effect. This provides the necessary flow characteristics.

One design of this type of valve is the ball valve with an alternative opening which, in a state of no pressure, permits the exchange of the elastic ring seat of the ball without removing the valve itself from the supply line (Figure 3). This device is particularly useful with large nominal diameters. It is a further requirement that the components for transmitting the motion and the drive unit must be solidly built and require no maintenance. For this reason, massive shaft bearings in the horizontal direction are needed. These are not exposed to the deposits of solid matter, and thus are not located at the deepest point. The resulting possible lateral arrangement of the driving mechanism

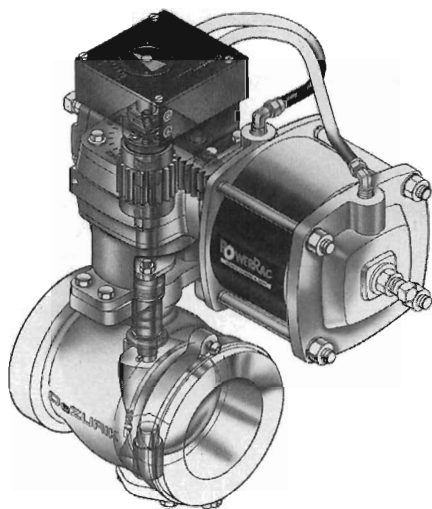


Figure 4. V-port ball control valve.

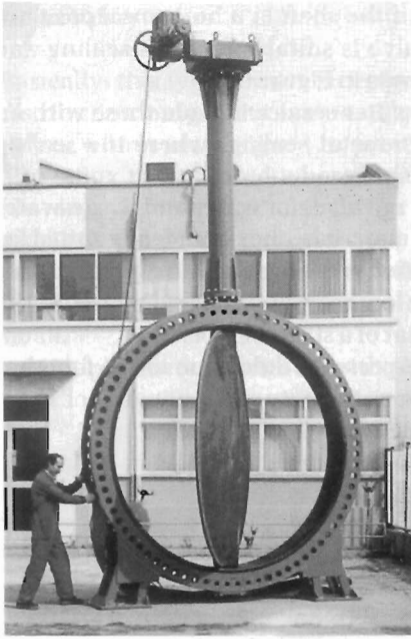
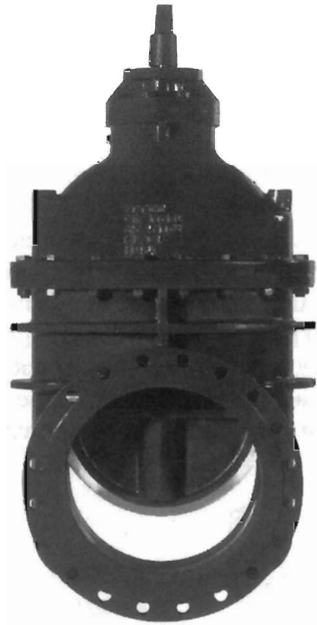


Figure 5. Centred-disk elastomer-lined butterfly valve for large pipe sizes.



Metal-seated gate valve for potable-water and dirty-water applications above and below ground.

results in a relatively low construction height. These construction considerations have led to a durable and maintenance-free design of ball valves for the drinking-water supply using well-tested components at competitive prices.

The ball valve shown in Figure 4 is suitable for fibrous suspension applications as well as clean water. The construction of this valve makes it ideal for pulp and paper industry duties. The packing gland is investment cast 316 stainless steel with a standard PTFE packing.

In line with many developed control valves, the internal flow passages have been computer designed. Shut-off is to ANSI Class VI.

PTFE-seated butterfly valves with a double-offset design offer optimum valve life, particularly if they are used primarily for isolation and flow control.

The elastomer-lined centred-disk butterfly valve (Figure 5) has been specially designed for large pipe sizes—1100–3200 mm (44–128 in). This type of valve is mainly used in water supply, water treatment and electric power stations.

In terms of standard performance and durability, rubber-lined butterfly valves still command a significant position in the water and waste-water industry, although PTFE is the outstanding performer. Where a rubber lining is bonded to the body of the valve, making it an integral part of the body, the valve body is not in contact with the medium so corrosion between body and

lining is not an issue. When installed with the shaft in a horizontal position, the valve is self-cleaning. This type of valve is suitable for both sealing and control functions. A typical example is shown in Figure 6.

Other types of butterfly valves used in water service include those with an inclined-cone sealing system for metal-to-metal sealing, where the sealing system is completely integrated within the body.

Gate valves (Figure 7) are still the primary valves for water and waste-water service. Manufactured in a wide range of materials, they are ideally suited for on-off duties. The valves have knife gates, wedge gates, and parallel face gates. Generally, these valves have a very low resistance to flow which, in the case of parallel-gate valves, approaches that of a straight pipe.

They are also used for duties with high-pressure fluids due to the fact that upstream pressure assists the sealing between the gate and seat.

Gate valves tend to be handwheel operated for water service.

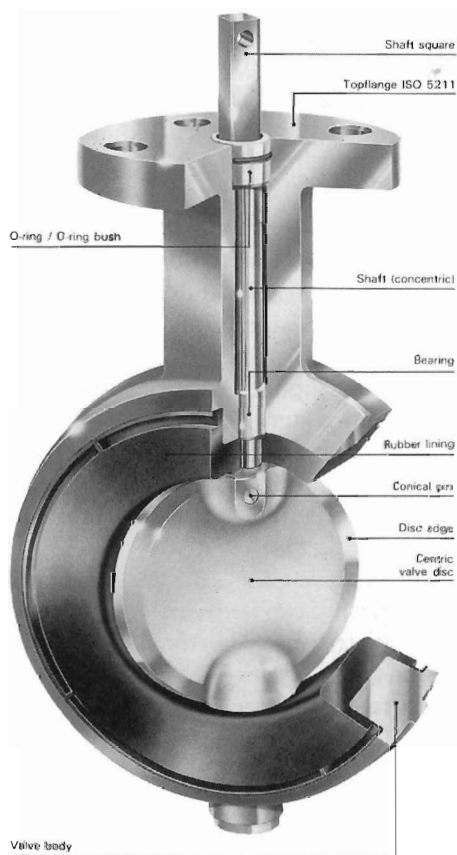


Figure 6. Rubber-lined butterfly valve.

Reflux valves

Basically, this type of valve is designed for water-works duties such as normal distribution on gravity mains. Certain types include non-slam recoil reflux valves which are designed to prevent flutter at high velocities, and rocking disc reflux valves, used where pipeline dimensions are in excess of 600 mm (24 in). Some reflux valves have outside weighted levers and heavy proportioned doors to provide non-slam characteristics and assist closing. A particular type of water-works reflux valve is the multiple-door reflux valve suitable for large-diameter pump or suction mains, where flow velocity is small.

Multiple doors combined with the large-area diaphragm provide a lower head loss against the valve than is possible with the valves of single door

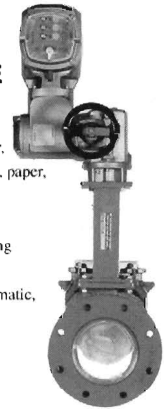
STAINLESS STEEL KNIFE GATE VALVES

- Major markets: Paper, chemical, mining, power, solids handling, OEM
- Size range 50mm to 2000mm
- Pressure rating PN10
- Raised seat face for positive seating
- Actuators: Manual, electric, pneumatic, hydraulic



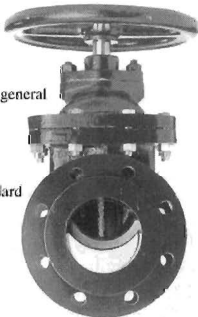
CAST IRON KNIFE GATE VALVES

- Major markets: Wastewater, water, chemical, mining, solids handling, paper, power, OEM
- Size range 80mm to 600mm
- Raised seat face for positive seating
- Pressure rating PN16
- Actuators: Manual, electric, pneumatic, hydraulic



WEDGE GATE VALVES

- Major markets: Wastewater, general industry
- Size range 50mm to 600mm
- Pressure rating PN16
- Meets BS5163 Type B Standard
- Gummel seat
- Actuators: Cup top, handwheel, electric motors



UNIVAL PORTED GATE VALVES

- Major markets: Paper, chemical, power, mining, solids handling
- Size range 80mm to 600mm
- Pressure rating: PN6 standard, PN10 on application
- Bi-directional
- One piece reinforced elastomer sleeve for sealing, abrasiv and corrosive services
- Full port opening
- Actuators: Manual, electric, pneumatic, hydraulic

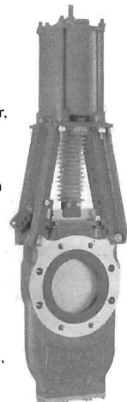


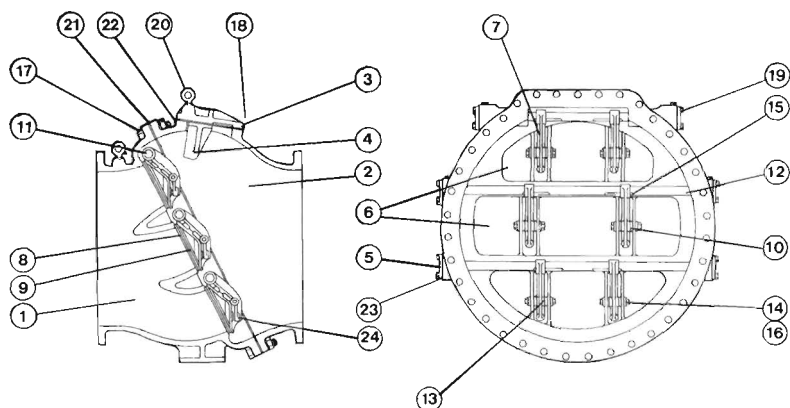
Figure 7. Examples of gate valves.

design. Also, the reduction of inertia reduces the risk of slamming as pump shut-down occurs. The design of the diaphragm inlet ports and body-contour shape largely avoids the action of cavitation. Valves of this type are typically suitable for velocities up to 3 m/s (10 ft/sec).

Non-return valves in water systems

The complexity of water-distribution network dynamics creates an unstable equilibrium of pressure and back-pressure, constantly modified by the user, leading to different appliances or collectors, some of which can be compared with veritable drains: retention vats in factories, sinks and their dishwasher, baths and their bath- or soap-water, washing machines, central-heating circuits with anti-scale additives, etc.

As a result, there is the possibility that water provided through the network can be polluted by waste-water leaving the consumers returning to the mains or passing from one consumer point to another without going through the mains (from one apartment to the other).



- | | |
|-----------------------------|---------------------------|
| 1. Body inlet | 14. Nuts and pins |
| 2. Body outlet | 15. Keys |
| 3. Cover | 16. Split pins |
| 4. Stop | 17. Stud bolts to body |
| 5. End caps | 18. Studs to cover |
| 6. Door | 19. Setscrews to end caps |
| 7. Rocker arms | 20. Eyebolts |
| 8. Faces to body | 21. Body joint |
| 9. Faces to door | 22. Cover joint |
| 10. Bushes to rocker arms | 23. End plate joint |
| 11. Bushes to end caps | 24. Buffers |
| 12. Spindles to rocker arms | 25. Air release plug |
| 13. Pins to rocker arms | |

Multiple-door reflux valve for large-diameter pump or suction mains.

This may be caused by:

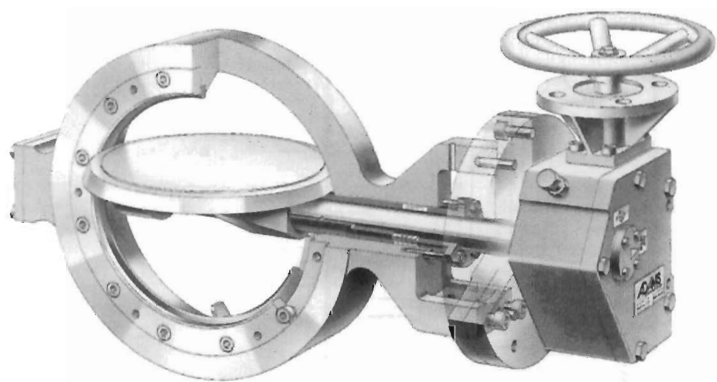
- (i) Depression on the mains: a considerable call for water (e.g. fire hydrants) or intervention in the main pipe (repair, new branch, breakage) can create a depression.
- (ii) Over-pressure at the consumer: all systems of high pressure, of course, but also all appliances for hot-water production, sanitary or otherwise, instantaneous or otherwise, can be the reason for this.
- (iii) Simultaneous appearance of low pressure on the mains and high pressure at the consumer.

Protection systems

Theoretically, it is imperative to implement those systems which prevent water returns due to these pressure disturbances. These systems should be automatic. They can be purely hydraulic, air-hydraulic or mechanical. The hydraulic systems are theoretically the most reliable but are often expensive



Non-return stop valve.



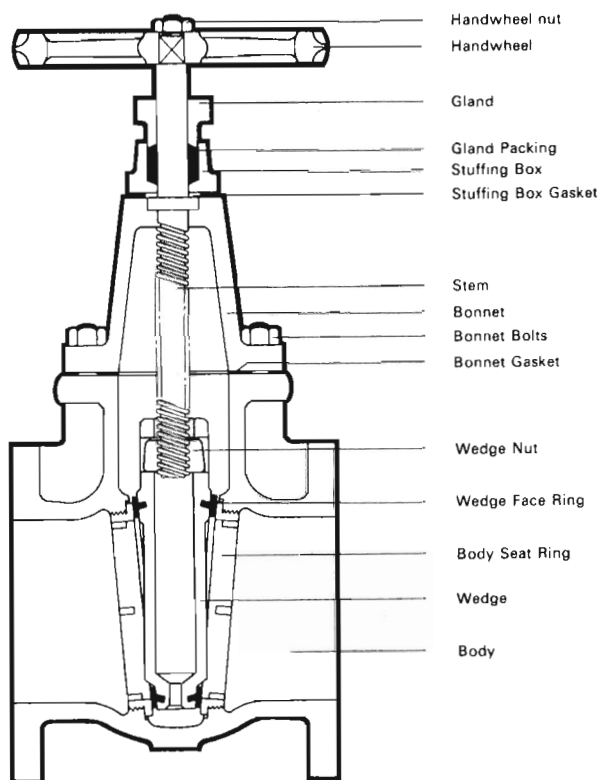
General purpose valve for water, sea-water and sewage.

or difficult to install. The mechanical systems are subject to doubts regarding function and longevity. Principal types of protection are summarised in Table 2.

In the presence of all these complex phenomena, and conscious of the necessity to organise a water distribution of quality and to protect it against the risks of pollution, numerous countries have brought about legislation which defines the measures to be taken (see Table 2).

The principle consists of defining the rules which a good general installation has to follow and the criteria of quality of the protection appliances to be incorporated, in order that the whole mains escapes the danger of pollution. To do this it is possible to use, individually or in combination, certain of the different hydraulic, air-hydraulic or mechanical systems examined later.

Most industrial countries have chosen to install non-return valves (NRV), taking into account the security-cost compromise and applying very strict design and control rules which considerably reduce the possible hazards connected with the mechanical character of the design.



Cast-iron gate valve.